








Modelling Financial Markets

Chapter 3: Factor Models and Asset Pricing



Daniel Traian PELE

Bucharest University of Economic Studies
IDA Institute Digital Assets
Blockchain Research Center
AI4EFin Artificial Intelligence for Energy Finance
Romanian Academy, Institute for Economic Forecasting
MSCA Digital Finance

Why Do Assets Have Different Average Returns?

What we will explain

- What explains expected returns? The Capital Asset Pricing Model (CAPM) and multifactor models
- How are exposures estimated? Betas, their uncertainty and shrinkage
- How are pricing restrictions tested? A joint test of the alphas, Fama–MacBeth regressions, model comparison
- Do the findings survive? Multiple testing, latent factors, replication

What you should be able to do

- estimate a beta with heteroskedasticity- and autocorrelation-robust errors and shrink it
- test the CAPM alphas jointly and estimate risk premia with robust errors
- judge a new factor against multiple-testing hurdles and replication evidence

Route through the Chapter

1. What Explains Expected Returns?
2. From Markowitz to the CAPM
3. Estimating Beta, with a quick check
4. Testing the CAPM, with a quick check
5. Multifactor Models
6. Fama–MacBeth Regressions, with a quick check
7. Discount Factors and Model Comparison
8. The Factor Zoo and Multiple Testing, with a quick check
9. Latent Factors: PCA and IPCA
10. Factor Investing
11. Case Study: Is There a Replication Crisis?
12. AI for Scientific Discovery
13. Conclusions

- **First pass**: Sections 1–6, 8 and 10, the results of Section 11 and the conclusions; answer each quick check before opening its answer in the Appendix
- **Deeper study**: slides marked *(deeper)* in the title (GMM, useless factors, discount factors, number of factors, IPCA, the Bayesian hierarchy of the case study) and the Appendix, reached through the buttons on the slides
- **Practice**: each method has a Seminar 3 problem, named and linked on the slide that uses it
- Section 12 prepares the course project

What Is a Factor Model?

- A **factor model** explains the return of asset i by a few common variables $f_{1,t}, \ldots, f_{K,t}$, the **factors**

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_{i,1} f_{1,t} + \cdots + \beta_{i,K} f_{K,t} + \varepsilon_{i,t}$$

- $R_{i,t}$: return of asset i in period t; $R_{f,t}$: risk-free rate; K: number of factors; α_i: average return not explained by the factors, 0 if they price the asset
- $\beta_{i,k}$: **factor loading** (exposure) of asset i to factor k; $\varepsilon_{i,t}$: asset-specific (idiosyncratic) shock
- Two uses in this chapter
  - Explain an asset: the technology ETF (exchange-traded fund) XLK moves 1.28 times the US market (Section 3)
  - Evaluate a fund: is its α_i positive once its exposures are priced (Sections 4 and 10)?
- Also: cost of capital, risk management (N assets need $N(N+1)/2$ covariances, K factors far fewer), portfolio construction (Campbell, Lo & MacKinlay (1997), ch. 5–6)
- Key message: expected returns should be explained by **exposures to priced risks**, not by the name of the asset

Which Data Test the Models: US Factors, ETFs and the BVB

Series	Content	Period
Fama–French factors	market, size, value, profitability, investment	1963–2026
Momentum factor	winners minus losers	1963–2026
25 size × B/M portfolios	US stocks, two sorts	1963–2026
US sector ETFs	S&P 500 sectors: 9 from 1999, 11 from 2018	1999–2026
Factor ETFs	momentum, value, quality, min. volatility, size	2013–2026
BVB blue chips, BET	11 stocks and their index	2015–2026

- Sector ETF tickers: XLB materials, XLE energy, XLF financials, XLI industrials, XLK technology, XLP consumer staples, XLU utilities, XLV health care, XLY consumer discretionary; from 2018 also XLRE real estate and XLC communication services

- Factors and portfolios: Kenneth French Data Library, in percent (divide by 100), monthly to July 2026 (the momentum file of the vintage used runs to August 2026); ETFs and stocks: daily adjusted closing prices (BET: closing values of the price index), to 18 Sep 2026
- **Excess return** $R^e_{i,t} = R_{i,t} - R_{f,t}$; $R_{f,t}$: one-month Treasury bill rate (RF in the French data); **Mkt-RF**: excess return of the value-weighted US stock market in the same data (**value-weighted**: weights proportional to market capitalisation)
- **B/M**: book value of equity over its market value; **BVB**: Bursa de Valori București, the Bucharest Stock Exchange; **BET**: its benchmark index

Simple or Log Returns?

- **Simple return** (the percentage price change): $R_{i,t} = P_{i,t}/P_{i,t-1} - 1$
  - $P_{i,t}$: adjusted price (dividends reinvested) of asset i at the end of period t
- **Log return**: $r_{i,t} = \ln(P_{i,t}/P_{i,t-1}) = \ln(1 + R_{i,t})$
- Example: the price rises from 100 to 110, so $R = 10\%$, but $r = \ln 1.1 = 9.53\%$
- If the price falls back to 100: $R = -9.09\%$ and $r = -9.53\%$, so the two log returns sum to 0
- Log returns add over time, simple returns add across assets: $R_{p,t} = \sum_i w_i R_{i,t}$ (Chapter 0)
- Hence regressions on the French factors use simple returns: the factors are portfolio returns
- Rolling-window, BVB and principal-component analyses (Section 9) use daily log returns
- The BVB cleaning rule $|r_t| > 0.5$ means a daily rise above 64.9% ($e^{0.5} - 1$) or a fall above 39.3% ($1 - e^{-0.5}$)
- BVB: daily $R_{f,t}$ treated as constant; it cancels in covariances, so betas are unchanged

Mean–Variance and the Tangency Portfolio

- Markowitz (1952): with weights w, expected excess returns $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$, the frontier solves

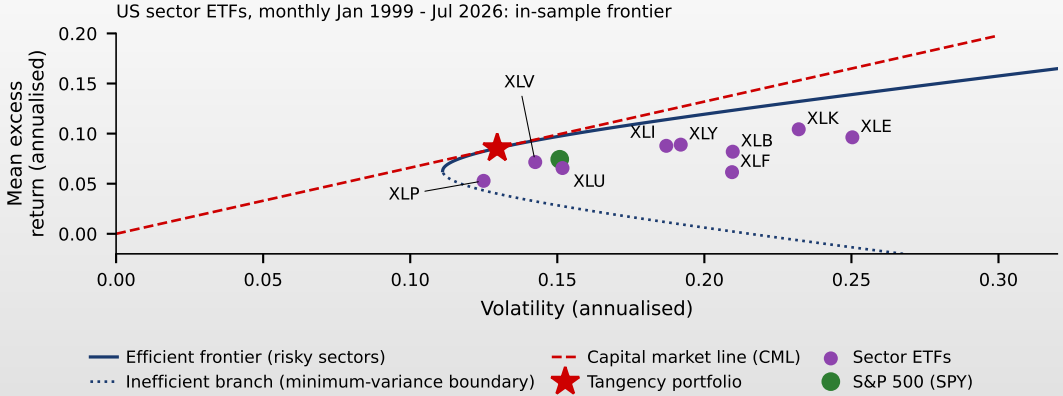
$$\min_{\mathrm{w}} \ \mathrm{w}^{\top}\boldsymbol{\Sigma}\,\mathrm{w} \quad \text{s.t.} \quad \mathrm{w}^{\top}\boldsymbol{\mu} = \mu^{*},\ \mathrm{w}^{\top}1 = 1$$

  - in words: the lowest variance for a target mean μ^{*}; 1: a vector of ones, so the weights sum to 1; $^{\top}$: transpose
- **Sharpe ratio** $SR_p = \mathbb{E}[R_p - R_f]/\sigma_p$, the excess return per unit of volatility; with a risk-free asset the **tangency portfolio** maximises this ratio when the covariance is positive definite and the denominator below is positive

$$\mathrm{w}_{\mathrm{tan}} = \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu} \,/\, 1^{\top}\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}$$

  - R_p, σ_p: return and volatility of portfolio p; the denominator rescales $\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}$ so that the weights sum to 1
- **CML** (Capital Market Line): mixes of R_f and the tangency portfolio, $\mu_p = SR_{\mathrm{tan}}\,\sigma_p$; negative weights are **short positions**
- From the tangency portfolio to the beta pricing equation: Seminar 3, problem A2; estimation error and shrinkage: Chapter 4

The Efficient Frontier of US Sectors (1/2)



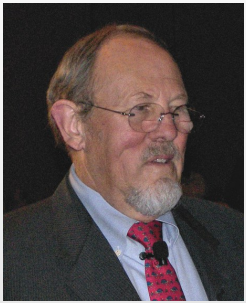
MFM_ch3_frontier

The Efficient Frontier of US Sectors (2/2)

- Monthly excess returns of 9 sector ETFs, January 1999 – July 2026 ($T = 331$ months)
- Tangency portfolio: mean 8.55%, volatility 12.96%, Sharpe ratio 0.66
- SPY (the S&P 500 ETF, the market proxy): mean 7.44%, volatility 15.08%, Sharpe ratio $7.44/15.08 = 0.49$: here the market is **not** on the frontier, although the CAPM requires it to be
- The chart axes are annualised fractions: 0.10 on an axis means 10%
- Tangency weights: -0.37 in financials, -0.21 in materials, $+0.37$ in health care
- The frontier is computed **in-sample** from noisy means (Chapter 4): whether the gap to SPY exceeds noise is tested on What a CAPM Test Establishes (not rejected, $p = 0.85$)

MFM_ch3_frontier

The CAPM: Sharpe, Lintner, Black



William F. Sharpe, Nobel Prize in Economics 1990

Photo: Larry D. Moore (2007); CC BY 4.0; Wikimedia Commons

- The **CAPM** was formulated by Sharpe (1964) and Lintner (1965)
- If every investor holds a mean–variance efficient portfolio, all hold the same tangency portfolio, which must then be the **market portfolio**
- Expected excess return is proportional to **beta**

$$\mathbb{E}[R_i] - R_f = \beta_i \left(\mathbb{E}[R_m] - R_f\right), \qquad \beta_i = \frac{\mathrm{Cov}(R_i, R_m)}{\mathrm{Var}(R_m)}$$

- R_i, R_m: returns of asset i and of the market portfolio; $\mathbb{E}[R_m] - R_f$: **market risk premium**; β_i: sensitivity of asset i to the market, $\beta_i > 1$ amplifies market moves
- Black (1972): a version without a risk-free asset, the zero-beta CAPM (Section 4)
- Sharpe shared the 1990 Nobel Prize with Markowitz and Miller

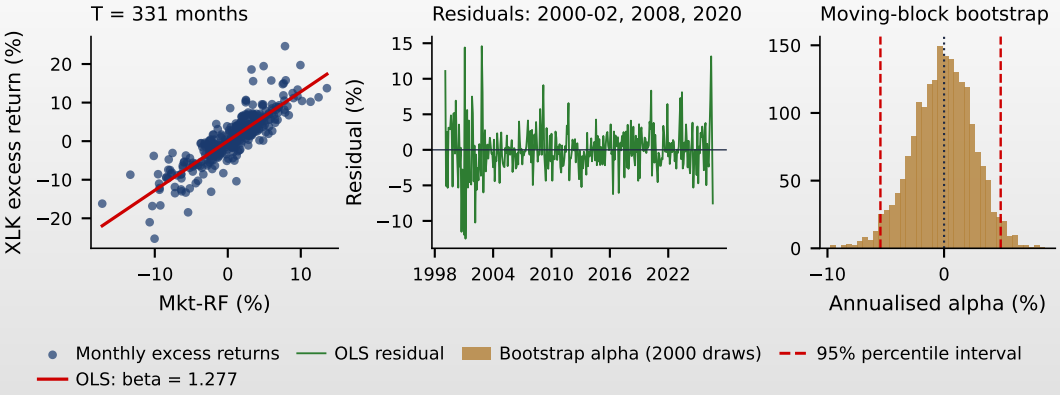
What the CAPM Implies

- Main assumptions
  - Mean–variance investors with identical beliefs, one period
  - Unlimited borrowing and lending at R_f; no taxes or trading costs
- Only beta is priced: no other characteristic should predict returns; idiosyncratic risk earns nothing
- **SML** (Security Market Line): the testable statement of the CAPM (Section 4)
  - every asset lies on the line (beta, mean), with slope equal to the market premium
  - the CML contains only efficient portfolios
- Cost of capital from the CAPM needs an estimated beta and an estimated premium: the error of the premium dominates
  - XLK, annual: premium s.e. $\approx \sigma_m/\sqrt{\text{years}} = 15.4\%/\sqrt{63} = 1.9\%$ (σ_m: volatility of Mkt-RF, 1963–2026), times $\hat{\beta} = 1.28$: 2.5%; beta s.e. times premium: $0.082 \times 7.2\% = 0.6\%$
- Which portfolio is "the market"? The answer decides what a test of the CAPM can establish (Roll's critique, Section 4)

The Market Model, HAC Errors and Estimation Choices

- Market model $R^e_{i,t} = \alpha_i + \beta_i R^e_{m,t} + \varepsilon_{i,t}$; **OLS** (Ordinary Least Squares): $\hat{\beta}_i = \widehat{\mathrm{Cov}}(R^e_i, R^e_m)/\widehat{\mathrm{Var}}(R^e_m)$
- Variance split $\sigma_i^2 = \beta_i^2\sigma_m^2 + \sigma^2_{\varepsilon,i}$: systematic (priced under the CAPM) plus idiosyncratic (diversifiable); R^2 is the systematic share
  - σ_i^2, σ_m^2: variances of the asset and of the market; $\sigma^2_{\varepsilon,i}$: residual variance; $R^2 = \beta_i^2\sigma_m^2/\sigma_i^2$
- Volatility clustering (Chapter 1) makes residuals heteroskedastic: use **HAC** (heteroskedasticity and autocorrelation consistent) errors (Newey & West (1987))
  - Bartlett weights and $L = \lfloor 4(T/100)^{2/9} \rfloor$ lags (Newey & West (1994)); T: number of periods, $\lfloor \cdot \rfloor$: integer part
  - $T = 331$: $L = 5$; $T = 757$: $L = 6$; the point estimates are those of OLS
- Choices to report with every beta
  - Frequency and window: 3–5 years of monthly or 1–2 years of daily data; daily data suffer from asynchronous trading
  - Market proxy: SPY (S&P 500) or Mkt-RF for US assets, BET for Romanian stocks
  - Data cleaning: a scale error in the first records creates a fake return (DIGI: the closes of 17–22 May 2017 are about 725 times those from 23 May on; daily log return -6.6)

Worked Example: Beta of the Technology ETF (1/2)



MFM_ch3_seminar

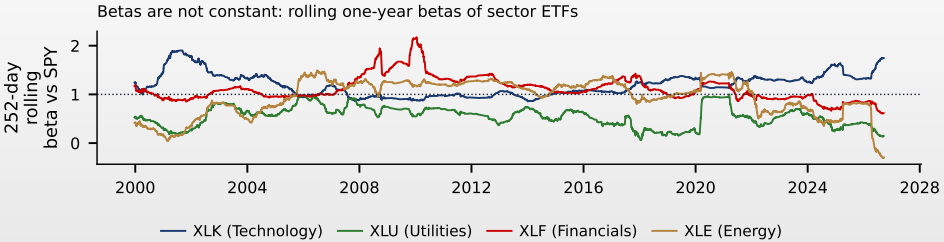
Worked Example: Beta of the Technology ETF (2/2)

		Standard error		
	Estimate	**Classical**	**White**	**Newey–West**
α (annual)	0.33%	2.27%	2.32%	2.45%
β	1.277	0.041	0.054	0.082

- XLK on Mkt-RF, monthly 1999–2026, $T = 331$, $R^2 = 0.742$
- **Classical**: errors assumed homoskedastic and serially uncorrelated; **White**: heteroskedasticity-consistent (White (1980)); **Newey–West**: also robust to autocorrelation, $L = 5$ lags
- Robust standard error (s.e.) of β: twice the classical one; heteroskedasticity explains part (White), serial correlation of $f_t\varepsilon_t$ the rest
- $t(\alpha) = 0.33/2.45 = 0.13$; 95% bootstrap interval for α: $[-5.5\%, 4.9\%]$
  - moving-block bootstrap of the monthly pairs (R^e_t, f_t): 12-month blocks keep the serial dependence; 2000 draws, seed 42 (Seminar 3, problem B1); bootstrap s.e. 2.61%, so slightly wider than $\pm 1.96 \times 2.45\%$

MFM_ch3_seminar

Betas Are Not Constant



- OLS beta on the last 252 daily log returns, recomputed every day
- Financials (XLF): from 0.60 to 2.17; the peak follows the 2008–2009 banking crisis
- Technology (XLK): from 0.86 to 1.90; high in the dot-com years, 1.75 at the end
- Energy (XLE): from −0.31 to 1.49; negative at the end, when oil moved against the market
- Utilities (XLU): from 0.06 to 1.00, the classic defensive sector
- A full-sample beta hides these regimes; conditional models: Chapters 5–6

MFM_ch3_rolling_beta

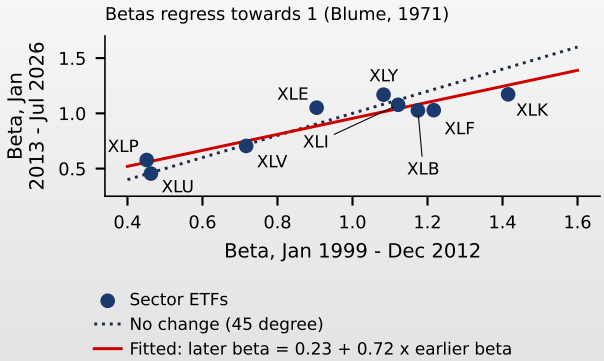
Shrinking Betas: Blume and Vasicek

- **Regression to the mean**: extreme estimated betas are partly noise and move towards 1 later
- Blume (1971): regress later betas on earlier betas; the practitioner rule is $\beta^{\text{Blume}} = 0.33 + 0.67\,\hat{\beta}$
- Vasicek (1973): Bayesian average of the estimate and the cross-sectional prior mean $\bar{\beta}$

$$\beta_i^V = w_i \hat{\beta}_i + (1 - w_i)\bar{\beta}, \qquad w_i = \frac{\sigma_\beta^2}{\sigma_\beta^2 + \text{se}(\hat{\beta}_i)^2}$$

- w_i: the weight of the estimate, larger when $\hat{\beta}_i$ is precise (small se); $\text{se}(\hat{\beta}_i)$: its standard error
- σ_β^2 is the variance of the **true** betas (Var_{cs}: variance across assets; $\overline{\text{se}^2}$: mean squared standard error): $\text{Var}_{cs}(\hat{\beta}) = \sigma_\beta^2 + \overline{\text{se}^2}$, so the empirical-Bayes prior is $\sigma_\beta^2 = \text{Var}_{cs}(\hat{\beta}) - \overline{\text{se}^2}$ (posterior derivation: Seminar 3, problem A3)
- XLK on SPY (not Mkt-RF, hence not the 1.28 of the worked example), monthly 1999–2012, prior from the 9 sectors: $\hat{\beta} = 1.415$ (Newey–West s.e. 0.139), $\bar{\beta} = 0.949$, $\sigma_\beta^2 = 0.116 - 0.008 = 0.108$, $w = 0.849$ from $0.1082/(0.1082 + 0.139^2)$ (unrounded inputs)
  - $\beta^V = 0.849 \times 1.415 + 0.151 \times 0.949 = 1.344$; Blume 1.278; beta estimated on 2013–2026: 1.172

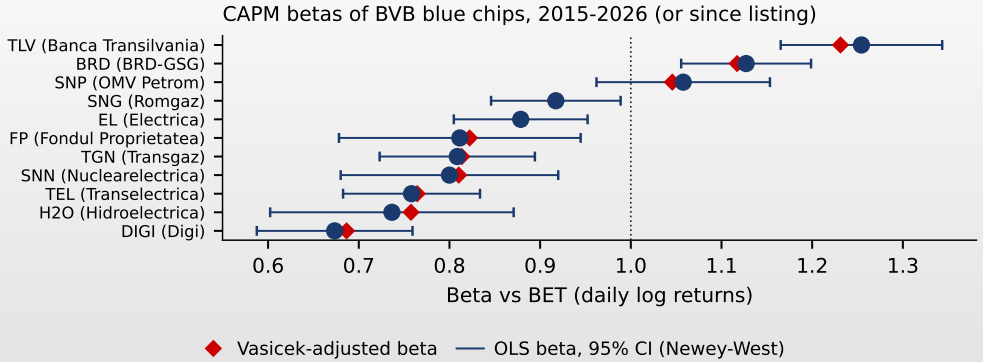
Do Betas Regress Towards One?



- Monthly betas of 9 sector ETFs on SPY: 1999–2012 (horizontal) vs 2013–2026 (vertical)
- Fitted line $\hat{\beta}_{\textbf{later}} = 0.23 + 0.72\,\hat{\beta}_{\textbf{earlier}}$, very close to Blume's $0.33 + 0.67\,\hat{\beta}$
- **RMSE** of the forecast of later betas, $\sqrt{\frac{1}{N}\sum_i (\text{forecast}_i - \hat{\beta}_{i,\textbf{later}})^2}$: raw 0.135, Blume 0.109, Vasicek 0.114
- Shrinkage cuts the forecast error by 15–20%; with 9 sectors the lesson is qualitative

MFM_ch3_beta_shrink

CAPM Betas on the Bucharest Stock Exchange (1/2)



Legend: OLS beta with its 95% confidence interval (CI), Newey–West errors; Vasicek-adjusted beta

MFM_ch3_bvb_betas

CAPM Betas on the Bucharest Stock Exchange (2/2)

- Daily log returns, regressed on BET, from 6 January 2015 or the listing date (Digi, DIGI, 2017; Hidroelectrica, H2O, 2023); days with $|r_t| > 0.5$ removed (one for DIGI, two for Fondul Proprietatea, FP)
- Banks are the most market-sensitive: Banca Transilvania (TLV) 1.25, BRD–Groupe Société Générale (BRD) 1.13; OMV Petrom 1.06; Transelectrica 0.76, Digi 0.67
- R^2: 0.59 for TLV, only 0.21 for DIGI; large index weights raise R^2 mechanically
- Vasicek changes little: the daily estimates are precise (se $\approx$ 0.04–0.07); stale prices bias daily betas of less liquid stocks (Seminar 3, problem B4)
- BET is a price index, the stock prices are dividend-adjusted: an alpha would absorb beta times the index dividend yield; the betas, driven by daily co-movement, should change little (the BET-TR comparison is not tested here)

MFM_ch3_bvb_betas

Quick Check: A Beta, Its Error and Its Shrinkage

- A stock, 120 monthly returns: $\hat{\beta} = 0.90$, classical s.e. 0.05, Newey–West s.e. 0.09
  1. Give both 95% intervals. Is the stock less risky than the market under each?
  2. Which interval do you report, and why does the point estimate not change?
- A cross section of betas has $\mathrm{Var}_{cs}(\hat{\beta}) = 0.10$, $\overline{\mathrm{se}^2} = 0.02$ and mean $\bar{\beta} = 1.0$
  3. Compute σ_β^2 and the Vasicek beta of an asset with $\hat{\beta} = 1.6$, once with s.e. 0.2 and once with s.e. 0.1
  4. Which of the two is shrunk more, and why? Compare both with the Blume beta
- Answers: ▸ Appendix

From One Alpha to the GRS Test (1/2)

- **Jensen's alpha** (Jensen (1968)): the market-model intercept α_i
  - The CAPM implies $\alpha_i = 0$ for every asset
- A separate t-test for each asset: XLK has $t = 0.13$
  - False rejections accumulate when the assets are many (Section 8)
- **GRS** (Gibbons, Ross & Shanken (1989)): a joint test of $H_0 : \alpha_1 = \cdots = \alpha_N = 0$

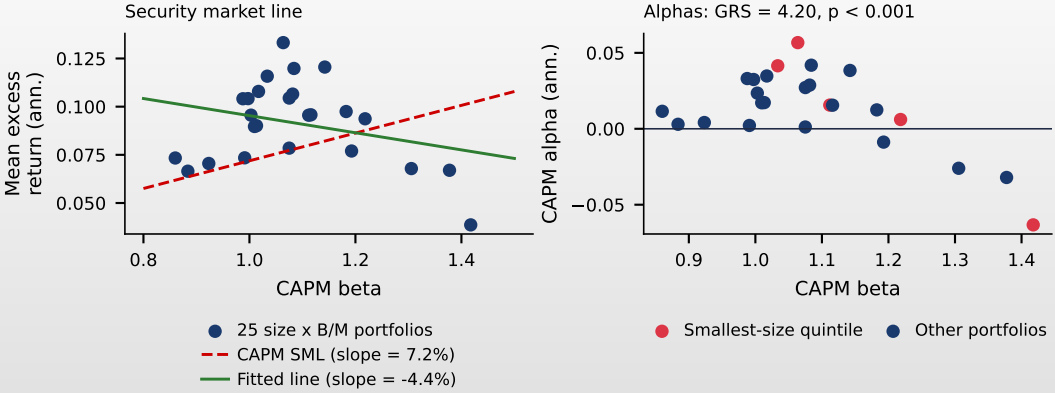
$$W = \frac{T - N - 1}{N} \, \frac{\hat{\boldsymbol{\alpha}}^\top \hat{\boldsymbol{\Sigma}}^{-1} \hat{\boldsymbol{\alpha}}}{1 + \hat{\mu}_m^2 / \hat{\sigma}_m^2} \;\sim\; F(N,\, T - N - 1)$$

- Notation
  - N: number of test assets; T: number of months; $\hat{\boldsymbol{\alpha}}$: the N estimated alphas
  - $\hat{\boldsymbol{\Sigma}}$: maximum-likelihood residual covariance (divided by T)
  - $\hat{\mu}_m / \hat{\sigma}_m$: Sharpe ratio of the market

From One Alpha to the GRS Test (2/2)

- $\hat{\boldsymbol{\alpha}}^{\top}\hat{\boldsymbol{\Sigma}}^{-1}\hat{\boldsymbol{\alpha}}$: the alphas squared and weighted by the inverse residual covariance
- Decision: reject H_0 when W exceeds the 95% quantile of $F(N, T-N-1)$; F: Fisher distribution
- The distribution is exact, conditional on the factors
  - errors i.i.d. (independent and identically distributed) multivariate Normal, independent of the factors
  - zero mean and constant nonsingular covariance
- Large W: the market is far from the mean–variance frontier spanned by the market and the test assets

The Security Market Line in the Data (1/2)



MFM_ch3_sml

The Security Market Line in the Data (2/2)

- Test assets: US stocks sorted each June into 5 size and 5 B/M quintiles, 25 value-weighted portfolios (Fama & French (1993)), July 1963 – July 2026, $T = 757$
- The CAPM line has slope 7.2% a year (chart axes in annual fractions: $0.05 = 5\%$); the fitted line has slope -4.4%: higher beta, **lower** average return
- Betas span only 0.86–1.42, average excess returns 3.9%–13.3%
- Small-value alpha $+5.7\%$, small-growth -6.3% a year; 10 of 25 alphas have $|t| > 1.96$
- GRS $= 4.20$, p-value 8×10^{-11}: the CAPM is rejected

MFM_ch3_sml

GRS as a Sharpe-Ratio Test

- Identity (Gibbons, Ross & Shanken (1989)): with $\hat{\boldsymbol{\Sigma}}$ and the moments estimated by maximum likelihood

$$\hat{\boldsymbol{\alpha}}^{\top}\hat{\boldsymbol{\Sigma}}^{-1}\hat{\boldsymbol{\alpha}} = \widehat{SR}^2(f,\mathrm{R}) - \widehat{SR}^2(f)$$

  - $\widehat{SR}^2(f,\mathrm{R})$: squared Sharpe ratio of the tangency portfolio of the factor and the N test assets; $\widehat{SR}^2(f)$: that of the factor alone
- So GRS measures the Sharpe-ratio gain from the test assets

$$W = \frac{T-N-1}{N}\cdot\frac{\widehat{SR}^2(f,\mathrm{R}) - \widehat{SR}^2(f)}{1+\widehat{SR}^2(f)}$$

- 25 portfolios, monthly Sharpe ratios: market 0.134 (unrounded 0.1344, $\times\sqrt{12}$: 0.47 a year); market plus portfolios 0.405 (1.40 a year)
- $W = 29.24 \times (0.405^2 - 0.134^2)/(1+0.134^2) = 4.20$, with $29.24 = (757-25-1)/25$; the gain expected under H_0 is far smaller (Appendix)
- Proof of the identity, and these numbers as its replication target: Seminar 3, problem A4

The Generalized Method of Moments in Brief (1/2) *(deeper)*

- A model implies $\mathbb{E}[\mathrm{g}_t(\boldsymbol{\theta})] = 0$ for a vector of moment functions g_t; **GMM** (Generalized Method of Moments, Hansen (1982)) in the exactly identified case chooses $\hat{\boldsymbol{\theta}}$ so that the sample mean is zero

$$\bar{\mathrm{g}}(\hat{\boldsymbol{\theta}}) = \tfrac{1}{T}\textstyle\sum_{t=1}^{T} \mathrm{g}_t(\hat{\boldsymbol{\theta}}) = 0$$

  - as many moments as parameters (exactly identified): the method of moments; with more moments, $\bar{\mathrm{g}}$ is made as small as possible in a weighted norm

- Sandwich variance in the exactly identified case

$$\mathrm{Var}(\hat{\boldsymbol{\theta}}) \approx \tfrac{1}{T}\,\mathrm{D}^{-1}\mathrm{S}\,\mathrm{D}^{-1\top}, \qquad \mathrm{D} = \mathbb{E}\big[\partial \mathrm{g}_t / \partial \boldsymbol{\theta}^{\top}\big], \qquad \mathrm{S} = \textstyle\sum_{j=-\infty}^{\infty} \mathrm{Cov}(\mathrm{g}_t, \mathrm{g}_{t-j})$$

  - D: Jacobian of the moments; S: their long-run variance, estimated by Newey–West (Section 3)

- Toy check, a mean: $g_t = R_t - \mu$ gives $D = -1$ and $\mathrm{Var}(\hat{\mu}) \approx S/T$, the HAC t-statistic of a mean factor return (Do the Premia Survive?)

- OLS: $\mathrm{g}_t = \mathrm{x}_t(y_t - \mathrm{x}_t^{\top}\boldsymbol{\theta})$ gives $\mathrm{D} = -\mathbb{E}[\mathrm{x}_t\mathrm{x}_t^{\top}]$: White errors when S has no lags, Newey–West errors otherwise (XLK table)

The Generalized Method of Moments in Brief (2/2) *(deeper)*

- **GRS as GMM**: parameters (α_i, β_i) for $i = 1, \dots, N$; $2N$ moments $\varepsilon_t \otimes (1, f_t)$, each residual times 1 and times the factor ($\otimes$: Kronecker product)
  - exactly identified, so $\hat{\boldsymbol{\theta}}$ is OLS asset by asset; only the variance is new
  - this ordering runs asset by asset, $(\alpha_1, \beta_1, \alpha_2, \beta_2, \dots)$, and $\mathrm{D} = -\mathrm{I}_N \otimes (\mathrm{Z}^\top \mathrm{Z}/T)$, with Z the $T \times 2$ matrix of rows $(1, f_t)$
  - Seminar 3, problem B7 stacks $(1, f_t)^\top \otimes \varepsilon_t$: the N alphas come first and $\mathrm{D} = -(\mathrm{Z}^\top \mathrm{Z}/T) \otimes \mathrm{I}_N$; same test, different bookkeeping (the sign of D cancels in the sandwich)
- $\hat{\mathrm{V}}_\alpha$: the rows and columns of the sandwich that belong to the alphas, with a HAC $\hat{\mathrm{S}}$
- Wald statistic $W_{\mathrm{GMM}} = \hat{\boldsymbol{\alpha}}^\top \hat{\mathrm{V}}_\alpha^{-1} \hat{\boldsymbol{\alpha}} \to \chi^2_N$ under H_0
- The same machinery gives the misspecification-robust premia errors (▸ Appendix) and estimates the discount-factor model of Section 7

When the GRS Assumptions Fail

- **Heavy tails and volatility clustering** (Chapter 1): the F distribution is no longer exact
  - Residual bootstrap under H_0: resample whole rows of residuals, rebuild returns with $\alpha = 0$ (Seminar 3, problem B7)
  - The bootstrap is robust to non-Normality only if the errors are i.i.d. and independent of the factors
  - GMM (Generalized Method of Moments) Wald test: moments $\varepsilon_t \otimes (1, f_t)$, each residual times 1 and times the factor ($\otimes$: Kronecker product; set-up and ordering: GMM in Brief)
  - $W_{\text{GMM}} = \hat{\boldsymbol{\alpha}}^\top \hat{\mathbf{V}}_\alpha^{-1} \hat{\boldsymbol{\alpha}} \to \chi^2_N$, $\hat{\mathbf{V}}_\alpha$: HAC estimate of $\text{Var}(\hat{\boldsymbol{\alpha}})$; robust to heteroskedasticity and autocorrelation
- **Rank condition**: $\hat{\boldsymbol{\Sigma}}$ has rank at most $T - 2$
  - GRS is not computable when $N \geq T - 1$, and its power falls well before that point
- Tests built for many assets: ▸ Appendix

A Flat SML: Black and Betting Against Beta



Fischer Black (1938–1995)

Photo: Dalmatine at English Wikipedia (2009 upload); public domain; Wikimedia Commons

- Black (1972): with restricted borrowing, beta pricing holds with a **zero-beta** rate above R_f in place of the risk-free rate
- Prediction: the SML is **flatter** than the CAPM line and starts at the zero-beta rate
- Frazzini & Pedersen (2014): constrained investors who want more risk buy high-beta assets instead of using leverage; these assets become expensive and earn less
- Their strategy, **BAB** (Betting Against Beta)

$$r_t^{\mathbf{BAB}} = \frac{R_{L,t} - R_{f,t}}{\beta_L} - \frac{R_{H,t} - R_{f,t}}{\beta_H}$$

- L: low-beta portfolio leveraged to beta 1; H: high-beta portfolio de-leveraged to beta 1; BAB has beta zero by construction and a positive average return in their data
- Per unit of capital: $1/\beta_L > 1$ in L, the difference borrowed at R_f; $1/\beta_H < 1$ in H, the rest lent at R_f; their excess returns are the two fractions above
- An unscaled low-minus-high-beta spread (Seminar 3, problem C2) is not beta-neutral: its mean mixes the premium with a negative market exposure

What a CAPM Test Establishes

- Roll (1977): the CAPM is a statement about the **true** market portfolio of all wealth; stock indices omit bonds, real estate, private firms and human capital
- Any efficient portfolio prices all assets exactly by beta: with R_f on the CML, with a zero-beta rate (Black (1972)) on the risky-asset frontier (Seminar 3, problem A2 Extended)
  - A CAPM test is a joint test of the model and of the efficiency of the proxy
- **Test assets matter**: 9 sector ETFs, January 1999 – July 2026 ($T = 331$): GRS $= 0.53$, $p = 0.85$, not rejected
  - The sectors of the frontier slide, with Mkt-RF as proxy (Sharpe ratio 0.51 a year, SPY 0.49): the in-sample gain 0.015 is below the 0.029 expected from noise (Appendix)
  - On the same 331 months the 25 size $\times$ B/M portfolios still reject: GRS $= 2.91$, $p = 9 \times 10^{-6}$
  - Sectors mix small and big, value and growth: they do not spread the characteristics that matter
- Failing to reject is weak evidence; evaluate models by how well they price assets with dispersion in expected returns

Quick Check: Reading a GRS Statistic

- One factor, $N = 4$ test assets, $T = 65$ months; monthly Sharpe ratios: factor alone 0.15, tangency portfolio of the factor and the 4 assets 0.25
  1. Compute W from the Sharpe-ratio form. The 95% quantile of $F(4, 60)$ is about 2.53: reject?
  2. Compute the squared Sharpe-ratio gain expected under H_0, $N(1 + \widehat{SR}^2(f))/(T - N - 3)$, and compare it with the observed gain
  3. Same Sharpe ratios, $T = 650$ months: recompute W (95% quantile of $F(4, 645)$: about 2.39). What changed, the gap or the evidence?
- Link to the chapter: the sector frontier shows a visible gap to SPY, yet GRS does not reject on 331 months
- Answers: Appendix

Arbitrage Pricing Theory

- **APT**: Arbitrage Pricing Theory, Ross (1976)
- Returns follow a K-factor model with small idiosyncratic risks

$$R_{i,t} = \mathbb{E}[R_i] + \boldsymbol{\beta}_i^\top \mathsf{f}_t + \varepsilon_{i,t}, \qquad \mathbb{E}[\mathsf{f}_t] = 0$$

- **No arbitrage** (no riskless profit without investment) implies, approximately

$$\mathbb{E}[R_i] - R_f = \boldsymbol{\beta}_i^\top \boldsymbol{\lambda}$$

- $\boldsymbol{\beta}_i$: the K loadings of asset i; f_t: the factor surprises, with mean 0; $\boldsymbol{\lambda}$: the **risk premia**, the extra expected return per unit of each exposure
- The theory does not say **which** factors: they must be found in the data

Building Factors: Long–Short Portfolios

size: NYSE median; B/M: 30% / 40% / 30%

	Low (growth)	Neutral	High (value)
Small	S_L	S_N	S_H
Big	B_L	B_N	B_H

$$\text{SMB} = \tfrac{1}{3}(S_L + S_N + S_H) - \tfrac{1}{3}(B_L + B_N + B_H)$$

$$\text{HML} = \tfrac{1}{2}(S_H + B_H) - \tfrac{1}{2}(S_L + B_L)$$

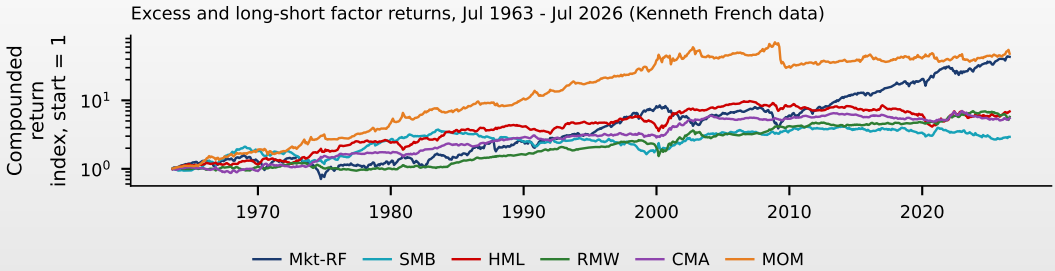
- A **characteristic-sorted factor** buys stocks with a high value of a characteristic and sells those with a low value
- Zero net investment: the return is a pure spread, $f_t = R_{\textbf{high},t} - R_{\textbf{low},t}$
- Fama–French 2×3 sort each June, accounting data of the previous year, value-weighted portfolios; the size breakpoint is the median NYSE (New York Stock Exchange) market capitalisation
  - **SMB** (Small Minus Big): small minus big stocks
  - **HML** (High Minus Low): high B/M (value) minus low B/M (growth)
- The same construction on profitability and investment gives RMW and CMA (next slide)

The Standard Factor Models

Model	Factors	Reference	Motivation
CAPM	Mkt-RF	Sharpe (1964)	equilibrium
FF3	Mkt-RF, SMB, HML	Fama & French (1993)	size and B/M effects (Fama & French (1992))
Carhart	FF3 + MOM	Carhart (1997)	fund evaluation
q-factor	market, size, investment, ROE	Hou, Xue & Zhang (2015)	firms' investment decisions
FF5	Mkt-RF, SMB, HML, RMW, CMA	Fama & French (2015)	dividend-discount model

- FF3 and FF5: the Fama–French three- and five-factor models; the FF3 regression is $R^e_{i,t} = \alpha_i + b_i\,\mathrm{MKT}_t + s_i\,\mathrm{SMB}_t + h_i\,\mathrm{HML}_t + \varepsilon_{i,t}$, with MKT = Mkt-RF and loadings b_i, s_i, h_i on the three factors; a good model leaves time-series alphas close to zero (GRS not rejected)
- Factors beyond FF3 (ROE: Return On Equity)
  - **RMW** (Robust Minus Weak): high minus low operating profitability
  - **CMA** (Conservative Minus Aggressive): low minus high asset growth
  - **MOM** (UMD, Up Minus Down): high minus low return from month $t-12$ to $t-2$ (the 12–1 signal); past winners keep winning over 3–12 months (Jegadeesh & Titman (1993))
- FF3 uses its own SMB (2×3 sorts on size and B/M); the FF5 SMB averages size spreads from three sorts, so FF3 is not nested in FF5

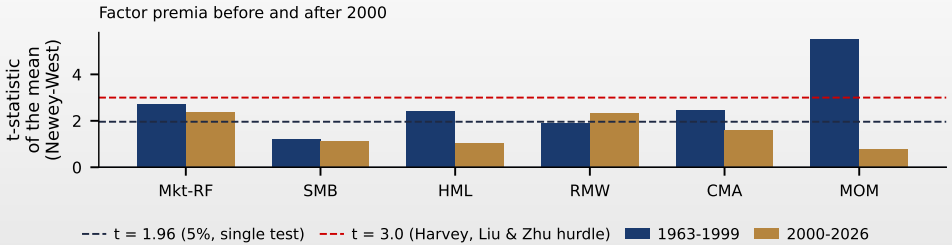
Six Factors over Sixty-Three Years



- Compounded monthly factor returns from 1, log scale: excess or long–short returns, so no interest on capital or on the collateral (the cash posted as margin for the short leg); the curves are not the wealth of a funded investment
- July 1963 – July 2026: Mkt-RF 43.2, MOM 48.0, HML 6.9, RMW 5.7, CMA 5.5, SMB 2.9
- Momentum peaked before 2009 and lost heavily in the 2009 rebound: a **momentum crash**
- The value factor was flat or losing from about 2007 to 2020
- Average annual returns: Mkt-RF 7.2%, MOM 7.3%, HML 3.6%, RMW 3.1%, CMA 3.0%, SMB 2.3%

MFM_ch3_factor_cum

Do the Premia Survive?



- t-statistic of the mean monthly factor return (Newey–West), 1963–1999 vs 2000–2026
- Momentum: $t = 5.53$ before 2000, 0.75 after; mean 10.7% vs 2.5% a year
- Value (HML): $t = 2.39$ then 1.02; mean 4.2% vs 2.8%
- Profitability (RMW) is the exception: $t = 1.90$ then 2.31; the market premium holds in both halves ($t = 2.70$ and 2.39)
- A calendar split is not a publication event: dating the decay needs the design of Section 12

MFM_ch3_factor_tstats

The Two Passes



Eugene Fama, Morgan Stanley–AFA (American Finance Association) Award, 2008

Photo: TonyTheTiger (2008); CC BY-SA 3.0; Wikimedia Commons

- Fama & MacBeth (1973) estimate the risk premia $\boldsymbol{\lambda}$ of $\mathbb{E}[R_i^e] = \lambda_{\mathbf{0}} + \boldsymbol{\beta}_i^\top \boldsymbol{\lambda}$ (Fama: Nobel Prize 2013)
- **Pass 1**: time series, one regression per asset
  - returns $T \times N$, factors $T \times K \Rightarrow$ betas $\hat{\mathsf{B}}$, $N \times K$

$$R_{i,t}^e = a_i + \boldsymbol{\beta}_i^\top \mathsf{f}_t + \varepsilon_{i,t}$$

- **Pass 2**: cross section, each month one regression of the N returns on $\mathsf{X} = [1, \hat{\mathsf{B}}]$

$$R_{i,t}^e = \lambda_{\mathbf{0},t} + \hat{\boldsymbol{\beta}}_i^\top \boldsymbol{\lambda}_t + u_{i,t}, \qquad \hat{\boldsymbol{\lambda}} = \tfrac{1}{T} \textstyle\sum_t \hat{\boldsymbol{\lambda}}_t$$

- $\lambda_{\mathbf{0}}$: pricing error common to all assets (in Black (1972): the zero-beta rate in excess of R_f), zero if the model is right; a_i: pass-1 intercept; $u_{i,t}$: pass-2 residual; $\hat{\boldsymbol{\lambda}}$: average of the monthly slopes
- The method allows factors that are **not** traded portfolios, and firm characteristics as regressors

Standard Errors for Risk Premia (1/2)

- **Fama–MacBeth**: the monthly $\hat{\lambda}_{k,t}$ form a sample of size T

$$\text{se}_{FM}(\hat{\lambda}_k) = s(\hat{\lambda}_{k,t})/\sqrt{T}$$

  - $s(\cdot)$: the sample standard deviation over the T months
  - robust to cross-sectional correlation of the residuals
  - if $\hat{\lambda}_{k,t}$ is autocorrelated, use Newey–West on this series
- **Misspecification**: the Fama–MacBeth, Newey–West and Shanken (next slide) errors assume a correct model
  - pricing errors $\mathrm{e} = \boldsymbol{\mu} - \mathrm{X}\boldsymbol{\gamma} = 0$; $\boldsymbol{\mu}$: mean excess returns, $\boldsymbol{\gamma} = (\lambda_0, \boldsymbol{\lambda}^\top)^\top$
  - when GRS rejects the model, the noise in $\hat{\mathrm{B}}$ multiplies e and adds variance (Kan, Robotti & Shanken (2013); GMM form, built as in GMM in Brief, in the ▸ Appendix)

Standard Errors for Risk Premia (2/2)

- **Errors in variables**: pass 2 uses $\hat{\beta}$, not β, so se_{FM} is too small
- Shanken (1992) scales the non-factor part of the variance by $(1 + c)$

$$\text{Var}_{\text{Sh}}(\hat{\boldsymbol{\lambda}}) = (1 + c)\left[\text{Var}_{FM}(\hat{\boldsymbol{\lambda}}) - \tfrac{1}{T}\boldsymbol{\Sigma}_f\right] + \tfrac{1}{T}\boldsymbol{\Sigma}_f, \qquad c = \hat{\boldsymbol{\lambda}}^\top \boldsymbol{\Sigma}_f^{-1}\hat{\boldsymbol{\lambda}}$$

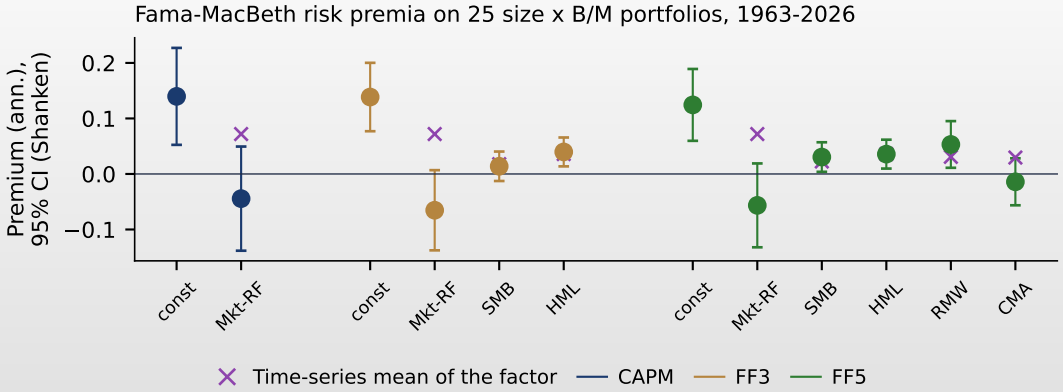
  - Var_{FM}: the Fama–MacBeth variance, $s^2(\hat{\boldsymbol{\lambda}}_t)/T$; $\boldsymbol{\Sigma}_f$: factor covariance
  - c: squared Sharpe ratio of the premia per period, 0.03 for FF3 here, larger for some non-traded factors
- Small c (traded factors, monthly data): the correction is small; large c: Fama–MacBeth t-statistics are overstated

Worked Example: Four Standard Errors for the Value Premium

FF3, HML	$\hat{\lambda}$	**s.e.**	t	**Significant at 5%?**
Fama–MacBeth	3.97%	1.33%	2.99	yes
Newey–West (6 lags)	3.97%	1.62%	2.45	yes
Shanken	3.97%	1.33%	2.99	yes
Kan–Robotti–Shanken	3.97%	1.32%	3.00	yes

- 25 size $\times$ B/M portfolios, 1963–2026; Shanken barely moves the error ($c = 0.03$ per month)
- Autocorrelation of the monthly slopes $\hat{\lambda}_t$ matters most here: Newey–West, 6 lags by the rule for $T = 757$
- FF5: for RMW, the misspecification-robust s.e. rises from 2.1% to 2.6%, and t falls from 2.5 to 2.1
- HML clears 1.96 under every choice; at the $t > 3$ hurdle of Section 8 it only touches the hurdle with Kan–Robotti–Shanken (3.00), sits just below with Fama–MacBeth and Shanken (2.99), and fails clearly with Newey–West (2.45)

Estimated Risk Premia (1/2)



MFM_ch3_fama_macbeth

Estimated Risk Premia (2/2)

- 25 size $\times$ B/M portfolios, 1963–2026; bars: 95% intervals with Shanken errors, in annual fractions ($0.1 = 10\%$); crosses: factor means
- CAPM: intercept 14.0% (s.e. 4.4%), market premium -4.4% (s.e. 4.8%)
- FF3: SMB 1.4% (1.4%), HML 4.0% (1.3%); FF5: RMW 5.3% (2.1%), CMA -1.4% (2.2%)
- For a traded factor, λ should equal its time-series mean; HML 4.0% vs 3.6%, close; market (FF3) -6.5% vs 7.2%, far apart
- Market betas span only 0.86–1.42 in the CAPM (cross-sectional standard deviation 0.14) and 0.89–1.14 in FF3 (0.07): λ_m is **weakly identified**, and a large intercept absorbs the level (Lewellen, Nagel & Shanken (2010))

MFM_ch3_fama_macbeth

Useless and Weak Factors *(deeper)*

- A **useless factor** g_t is independent of all returns: $\beta_g = 0$, so $\hat{\beta}_{i,g} = O_p(T^{-1/2})$, that is, of the order of $1/\sqrt{T}$: pure noise
- Pass 2 divides by this noise: $\hat{\lambda}_g$ does not converge to zero and its t-test over-rejects when the model is misspecified (Kan & Zhang (1999))
- Simulation on the 25 portfolios, CAPM plus an independent Normal factor, 2000 replications, nominal level 5%
  - $\lambda_g = 0$ rejected in 60% (Fama–MacBeth errors) and 27% (Shanken errors) of the replications
  - First-pass Wald test of $\beta_{i,g} = 0$ for all i: rejected in 6.7%; run it before pass 2
- **Weak identification**: betas close to zero (a weak factor) or nearly constant across assets (the market here)
  - identification-robust tests (Kleibergen (2009))
  - with many assets, consistency of pass 2 depends on factor strength (Anatolyev & Mikusheva (2022))

A Skeptical Appraisal: Adding 30 Industry Portfolios *(deeper)*

	OLS R^2		GLS R^2	
Model	25	25 + 30	25	25 + 30
CAPM	0.08	0.00	0.06	0.01
FF3	0.66	0.22	0.21	0.08
FF5	0.76	0.24	0.26	0.10

Monthly 1963–2026, free intercept and premia; replication target of Seminar 3, problem B6

- The 25 portfolios have a strong three-factor structure: any factors correlated with SMB and HML fit their means, so a high OLS R^2 is a low hurdle (Lewellen, Nagel & Shanken (2010))
- Three checks
  - Broaden the test assets: FF3 falls from 0.66 to 0.22 once 30 industries are added
  - Report the **GLS** (Generalised Least Squares) R^2, weighted like GRS by $\hat{V}^{-1}$, the inverse covariance matrix of the returns: never above 0.26
  - Impose $\lambda = \mathbb{E}[f]$ for traded factors, no intercept: FF3 on the 55 assets gives $R^2 = -0.27$
  - $R^2 = 1 - \text{SSE}/\text{SST}$ is negative when the imposed fit misses the means by more than their cross-sectional average does; OLS with a free intercept can always do at least as well as that average
- Stocks instead of portfolios: more dispersion in betas, larger errors in variables (Ang, Liu & Schwarz (2020))

Quick Check: Premia, Shanken and a Negative R^2

- A non-traded factor (for example, industrial-production growth) with volatility $\sigma_f = 1\%$ a month; $T = 400$ months; pass 2 gives $\hat{\lambda} = 0.5\%$ a month, with $s(\hat{\lambda}_t) = 3\%$
  1. Compute se_{FM} and the Fama–MacBeth t-statistic
  2. Compute c, the Shanken variance and the Shanken t-statistic. Does the factor clear the $t > 3$ hurdle?
  3. Why would c be tiny for a traded factor with $\sigma_f = 4.5\%$ and the same $\hat{\lambda}$?
- Three portfolios have mean excess returns 2%, 4%, 6% a year; a model with imposed premia and no intercept fits 6%, 4%, 2%
  4. Compute SST, SSE and R^2. Why can a free-intercept OLS fit never fall below 0?
- Answers: ▸ Appendix

Factors as a Stochastic Discount Factor *(deeper)*



Lars Peter Hansen, Nobel Prize in Economics 2013

Photo: Bengt Nyman (2013); CC BY 2.0; Wikimedia Commons

- **SDF** (Stochastic Discount Factor) m_t: prices all excess returns, $\mathbb{E}[m_t R^e_{i,t}] = 0$; m_t is high in bad states (the market falls), so an asset that pays off then is insurance and earns a low premium
- A linear factor model is an SDF, $m_t = 1 - \mathrm{b}^\top(\mathrm{f}_t - \mathbb{E}\mathrm{f})$; b: the SDF loadings, $\mathbb{E}\mathrm{f}$: the factor means
- Substitute: $\mathbb{E}[R^e_i] = \mathrm{Cov}(R^e_i, \mathrm{f}_t)^\top \mathrm{b} = \boldsymbol{\beta}_i^\top \boldsymbol{\lambda}$ with $\boldsymbol{\lambda} = \boldsymbol{\Sigma}_f \mathrm{b}$: the beta pricing model of Section 6
- $\mathrm{b} \neq 0$ asks whether a factor helps price the assets; $\boldsymbol{\lambda} \neq 0$ whether it is rewarded
- These moment conditions are estimated by GMM, Hansen (1982) (GMM in Brief: here $\mathrm{g}_t(\mathrm{b}) = m_t(\mathrm{b})\, \mathrm{R}^e_t$); survey of the discount-rate view: Cochrane (2011)

Comparing Models: Maximum Sharpe Ratios *(deeper)*

Max. Sharpe ratio (annual)	CAPM	FF3	Carhart	FF5	FF5 + MOM
1963–2026 ($T = 757$)	0.47	0.66	0.97	0.97	1.16
2000–2026 ($T = 319$)	0.48	0.54	0.67	0.97	1.02

- Barillas & Shanken (2018): judge a model by $SR^2(\mathrm{f}) = \boldsymbol{\mu}_f^\top \boldsymbol{\Sigma}_f^{-1} \boldsymbol{\mu}_f$ of its factors ($\boldsymbol{\mu}_f$, $\boldsymbol{\Sigma}_f$: means and covariance of the factors), the squared Sharpe ratio of their tangency portfolio; the test assets do not matter for the comparison
- Nested models: in-sample SR^2 can only rise, so test the alpha from the regression of the added factor on the others (spanning regression)
  - HML on the other five factors: $\alpha = 0.9\%$ a year ($t = 0.75$); since 2000 -1.7% ($t = -0.81$): no significant incremental contribution of HML, as in Fama & French (2015)
- Non-nested FF5 vs Carhart: difference in annual SR^2 0.00, 95% interval $[-0.57, 0.66]$; since 2000 0.50, $[-0.24, 1.48]$ (stationary bootstrap, Politis & Romano (1994); replication target of Seminar 3, problem B8)
- Another model-independent metric, the Hansen–Jagannathan distance: Appendix

The Factor Zoo

- Cochrane (2011) called the growing list of new factors a "zoo"
- Harvey, Liu & Zhu (2016) catalogue hundreds of published factors (316 in their list)
- Each was tested as if it were the only one: $|t| > 1.96$ at 5%
- With many tests, some pass by chance: **data mining**, also called p-hacking
- Their proposal: a new factor needs $|t| > 3.0$

Controlling False Discoveries

- **FDR** (False Discovery Rate): expected share of false rejections among all rejections
- **FWER** (Family-Wise Error Rate): probability of at least one false rejection (Chapter 2)
- Notation of the table: M: number of tests; α: significance level (not a pricing error); $p_{(1)} \leq \cdots \leq p_{(M)}$: sorted p-values; $H_{(j)}$: the hypothesis with the j-th smallest; **BH**: the Benjamini–Hochberg procedure

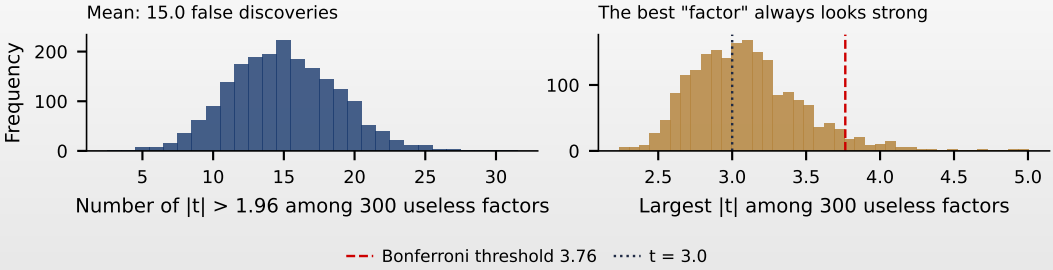
Procedure	**Controls**	**Reject $H_{(j)}$ if**	**Valid under**
Bonferroni	FWER	$p_{(j)} \leq \alpha/M$	any dependence
Holm (Holm (1979))	FWER	$p_{(k)} \leq \alpha/(M-k+1)$ for all $k \leq j$	any dependence
BH (Benjamini & Hochberg (1995))	FDR	$j \leq$ largest k with $p_{(k)} \leq k\alpha/M$	independence, positive dep.
BY (Benjamini & Yekutieli (2001))	FDR	as BH, thresholds divided by $\sum_{k=1}^{M} 1/k$	any dependence

- Positive dependence: the test statistics of the true nulls move up together (PRDS condition, positive regression dependence on a subset, Benjamini & Yekutieli (2001))
- FDR control suits a search over many candidate factors

Worked Example: Ten Candidate Factors

- Sorted p-values: 0.0004, 0.002, 0.006, 0.011, 0.018, 0.027, 0.041, 0.070, 0.15, 0.62; significance level $\alpha = 5\%$
- Naive ($p \leq 0.05$): **7** discoveries
- Bonferroni ($p \leq 0.005$): **2**
- Holm: $0.0004 \leq 0.005$, $0.002 \leq 0.0056$, $0.006 \leq 0.00625$, $0.011 > 0.0071$ stop: **3**
- BH thresholds $0.005j$: $p_{(6)} = 0.027 \leq 0.030$ is the last success: **6**
- Seminar 3, problem A1 repeats the procedure on a different vector, with a threshold chart

Simulation: 300 Useless Factors



- 2000 replications; each draws 300 independent t-statistics from the standard Normal distribution (no true premium)
- On average 15.0 useless factors pass $|t| > 1.96$; expected value $300 \times 0.05 = 15$
- On average 0.83 pass $|t| > 3$; at least one passes in 57% of the replications; median largest $|t|$: 3.06
- Bonferroni threshold for 300 tests: $|t| > 3.76$; hurdles rise with the number of tests actually tried, including unpublished ones

MFM_ch3_false_discoveries

Quick Check: Counting Discoveries

- Five candidate factors, sorted p-values 0.004, 0.012, 0.020, 0.030, 0.200; significance level $\alpha = 5\%$
  1. How many discoveries under the naive rule, Bonferroni, Holm and BH? Write down each threshold you compare with
  2. Under BY the BH thresholds are divided by $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5}$: how many discoveries remain?
  3. If all five nulls were true and the tests independent, what is the probability that at least one p-value falls below 0.05?
- Which count would you report for a search over many candidate factors, and which for a claim that **no** factor is spurious?
- Answers: ▸ Appendix

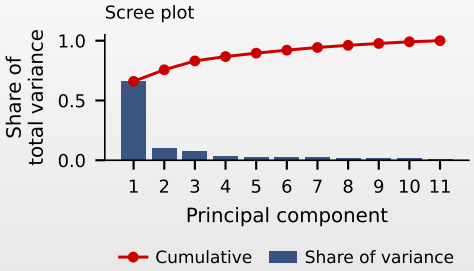
Principal Components in Brief

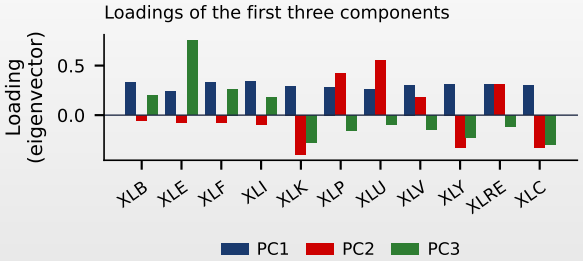
- **Latent factors**: estimated from returns alone, without naming them in advance
- **PCA** (Principal Component Analysis) of the correlation matrix C of standardised returns

$$\mathsf{C} = \mathbf{\Gamma}\mathsf{D}\mathbf{\Gamma}^\top, \qquad \mathsf{y}_t = \mathbf{\Gamma}^\top \mathsf{z}_t$$

  - z_t: standardised returns of day t; y_t: the components
  - $\mathbf{\Gamma}$: eigenvectors in columns; D: diagonal matrix of the eigenvalues ℓ_k
- Eigenvalues $\ell_1 \geq \cdots \geq \ell_N$ (ℓ, not the risk premium λ); eigenvectors are the **loadings**; component k explains ℓ_k/N of the variance
- Two assets with correlation ρ: $\ell = 1 \pm \rho$, eigenvectors $(1,1)/\sqrt{2}$ (the "market") and $(1,-1)/\sqrt{2}$ (the spread); in a crisis ρ rises and the first component dominates
- Textbook: Härdle & Simar (2019), chapter on principal components; N equicorrelated assets: Seminar 3, problem A8

PCA of the US Sector ETFs





- 11 sector ETFs, daily log returns, 20 June 2018 – 18 September 2026 ($T = 2073$ common days)
- The first principal component (PC1) explains 66.0%; all loadings positive (0.24–0.34); correlation with SPY 0.954: strongly market-related
- The second component, PC2 (9.7%): utilities $+0.55$, staples $+0.42$, real estate $+0.31$ vs technology -0.40, consumer discretionary -0.33: **defensive versus growth**, with no name given in advance
- The third, PC3 (7.4%), is dominated by energy; three components explain 83.1%

MFM_ch3_pca

How Many Factors? *(deeper)*

- Approximate factor model $x_{it} = \boldsymbol{\Lambda}_i^\top \mathrm{F}_t + e_{it}$, $N, T \to \infty$
  - x_{it}: standardised return; F_t: factors; $\boldsymbol{\Lambda}_i$: loadings; e_{it}: idiosyncratic part
  - $V(k)$: mean squared residual after the first k components; ℓ_k: the k-th largest eigenvalue
- Bai & Ng (2002): $\hat{k} = \arg\min_k \left[\ln V(k) + k \, \frac{N+T}{NT} \ln \min(N, T) \right]$ (criterion IC_{p2}); Ahn & Horenstein (2013): eigenvalue ratio $\hat{k} = \arg\max_k \ell_k / \ell_{k+1}$, no penalty to choose
- The rule of Kaiser (1960) ($\ell_k > 1$) has no asymptotic justification: the noise eigenvalues of a correlation matrix scatter around 1

Panel	N	T	Kaiser	Bai–Ng	Ahn–Horenstein
11 sector ETFs, daily 2018–2026	11	2073	2	5	1
100 size × B/M portfolios, monthly 1963–2026	95	757	3	4	1

- Standardised returns; maximum k: 5 for sectors, 8 for portfolios; 5 portfolios with missing months are dropped
- Ahn–Horenstein picks the dominant market factor (ratio 14.6); its next peak is at $k = 3$ (ratio 3.5), the FF3 structure
- Bai–Ng on 11 sectors hits the upper bound $k = 5$: the penalty is built for large N, so with small N the answer is fragile (▸ Appendix)

Limits of PCA for Asset Pricing

- Loadings are **constant**: PCA cannot follow a stock whose size or value changes
- Components explain **variance**, not necessarily **average returns**
- Individual stocks enter and exit the sample: a stock panel is unbalanced
- Component signs are arbitrary; rotations preserve principal components only within repeated-eigenvalue eigenspaces, while a latent-factor model is identified only up to rotation
- Risk-premium PCA and three-pass premia keep the PCA backbone but target pricing (▸ Appendix); PCA and factor analysis in finance: Franke, Härdle & Hafner (2019)

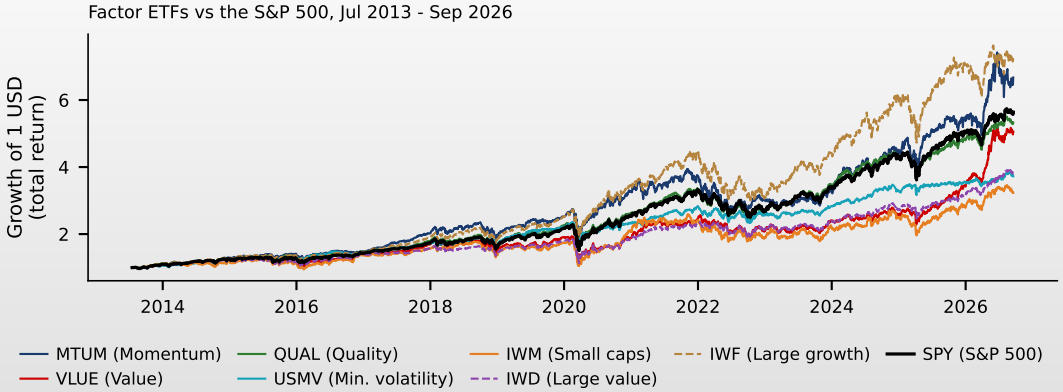
Instrumented PCA *(deeper)*

- **IPCA**: Instrumented Principal Component Analysis, Kelly, Pruitt & Su (2019)
- Loadings are functions of L observed characteristics $z_{i,t}$ (size, B/M, momentum, ...)

$$R^e_{i,t+1} = \alpha_{i,t} + \boldsymbol{\beta}_{i,t}^\top \mathbf{f}_{t+1} + \varepsilon_{i,t+1}, \qquad \boldsymbol{\beta}_{i,t}^\top = \mathbf{z}_{i,t}^\top \boldsymbol{\Gamma}_\beta$$

- $\boldsymbol{\Gamma}_\beta$ ($L \times K$) and the latent factors $\mathbf{f}_t$ are estimated jointly by alternating least squares
- Test: are characteristics priced through betas ("covariances"), or directly as alphas ($\alpha_{i,t} = \mathbf{z}_{i,t}^\top \boldsymbol{\Gamma}_\alpha$)?
- Result in their US stock data: a few IPCA factors price the cross section better than observable-factor models with as many factors, and $\boldsymbol{\Gamma}_\alpha$ is insignificant: "characteristics are covariances"; crypto-asset factors: Appendix

Factor ETFs versus the Market (1/2)



MFM_ch3_factor_etfs

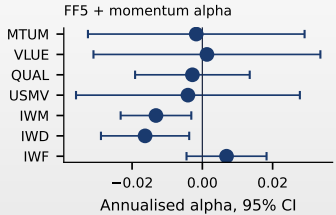
Factor ETFs versus the Market (2/2)

- **Smart beta**: rule-based long-only portfolios that tilt towards factors, sold as ETFs
  - they keep market exposure and need not replicate an academic long–short factor
- Thirteen years of one realisation: not evidence against the premia on its own

- **CAGR** (Compound Annual Growth Rate), $(V_T/V_0)^{1/n} - 1$ over n years, of 1 USD (US dollar) invested, dividends re-invested, from 18 July 2013
  - above SPY (14.0%): large growth (IWF) 16.3%, momentum (MTUM) 15.5%
  - below SPY: quality (QUAL) 13.5%, value (VLUE) 13.1%, large value (IWD) 10.7%, minimum volatility (USMV) 10.5%, small caps (IWM, Russell 2000 index) 9.3%
  - Russell 1000: the 1000 largest US stocks, with value (IWD) and growth (IWF) sub-indices; Russell 2000: the next 2000, smaller stocks (IWM)

MFM_ch3_factor_etfs

Alphas of Factor ETFs



Factor loadings

	Mkt-RF	SMB	HML	RMW	CMA	MOM
MTUM	1.03	-0.12	0.01	-0.22	-0.05	0.36
VLUE	0.98	0.08	0.34	-0.06	0.09	-0.03
QUAL	0.97	-0.08	-0.04	0.15	0.03	-0.04
USMV	0.74	-0.11	-0.02	0.20	0.21	-0.02
IWM	1.04	0.85	0.08	-0.10	-0.02	0.03
IWD	0.94	-0.04	0.29	0.09	0.19	-0.04
IWF	1.03	-0.11	-0.28	-0.03	-0.12	-0.00

- Daily regressions on FF5 + MOM, Newey–West errors, each ETF on its own trading calendar (2000–2026); chart alphas are annual fractions ($0.02 = 2\%$)
- The loadings match the labels: MTUM on MOM 0.36, VLUE on HML 0.34, IWM on SMB 0.85; IWF -0.28 on HML; USMV market beta 0.74
- Alphas close to zero: MTUM -0.2% ($t = -0.11$), QUAL -0.3%; Russell ETFs IWM -1.3% ($t = -2.57$), IWD -1.6% ($t = -2.54$), consistent with costs and index construction
- R^2 between 0.81 and 0.97: mostly repackaged factor exposure; replication target of Seminar 3, problem B11

MFM_ch3_etf_alphas

Timing, Crowding and Costs

- **Factor timing**: switching between factors based on signals (valuation spreads, recent returns)
  - Noisy premia and short histories make timing hard to verify out of sample (Chapter 8)
- **Crowding**: many investors hold the same factor positions
  - Premia can shrink (McLean & Pontiff (2016)), and crowded positions can amplify losses during forced unwinding; the 2009 momentum crash coincided with a sharp rebound of past losers (Daniel & Moskowitz (2016))
- **Implementability**: long–short factors ignore trading costs and short-selling fees; ETFs pay them

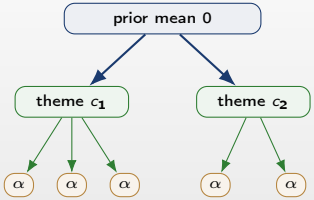
Do Published Factors Replicate?

- **The challenge** to factor research takes two forms (Jensen, Kelly & Pedersen (2023), Introduction)
  - No internal validity: results cannot be reproduced; Hou, Xue & Zhang (2020) replicate only 35% of their anomalies
  - No external validity: results come from testing many factors without controlling the FDR; Harvey, Liu & Zhu (2016) conclude that most claimed findings are likely false
- **Questions**
  - do published factors have significant CAPM alphas in the original data?
  - are these alphas significant out of sample and in other countries?
  - do they remain significant once multiple testing is taken into account?

Why Replication Matters for This Chapter

- **Why it matters**
  - if most factors are false discoveries, multifactor models and factor investing (Sections 5 and 10) rest on noise
- **A second view of the zoo** (Feng, Giglio & Xiu (2020), ▸ Appendix)
  - most new factors are redundant given the existing ones
  - Jensen, Kelly & Pedersen instead treat the zoo as correlated evidence on a few economic themes
- **Link to this chapter**
  - Jensen's alpha (Section 4) is the object tested
  - Benjamini–Yekutieli (Section 8) is the frequentist benchmark
  - Bayesian shrinkage extends the Vasicek beta (Section 3) to alphas

153 Factors Grouped into 13 Themes *(deeper)*



each factor's alpha borrows strength from the other factors of its theme

- **Data** (Section II): 153 factors in 93 countries
  - terciles of each characteristic, capped value weights
  - long the side with the higher expected return in the original paper
  - each factor scaled to a 10% annual idiosyncratic volatility
- **Themes**: 13 clusters of US factors
  - hierarchical clustering on the correlation of CAPM residuals
- **Model** (Section I): $f_{i,t} = \alpha_i + \beta_i r^m_t + \varepsilon_{i,t}$
  - $f_{i,t}$: return of factor i; r^m_t: market excess return
  - alphas $\alpha_i = c_j + s_n + w_i$: theme, signal and factor components, prior mean 0

Shrinking Alphas and the Replication Rate *(deeper)*

- One factor: the posterior mean shrinks the estimated alpha (eq. 5)

$$\mathbb{E}(\alpha \mid \hat{\alpha}) = \kappa \hat{\alpha}, \qquad \kappa = \frac{1}{1 + \sigma^2/(\tau^2 T)}$$

  - σ: residual volatility; τ: dispersion of the true alphas; T: number of months
  - short, noisy samples are shrunk more, as in Vasicek (Section 3): σ^2/T plays the role of $\text{se}(\hat{\beta}_i)^2$ and τ^2 that of σ_β^2
- **Replication rate**: share of the factors significant in the original studies whose alpha is significant in the new data
  - Bayesian discovery: posterior mean / posterior s.d. ≥ 1.96
- Full hierarchy and estimation: ▸ Appendix

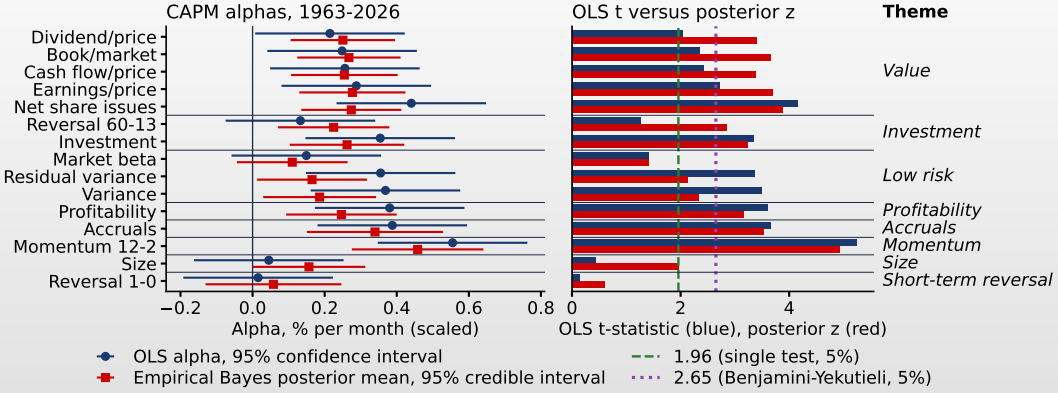
Most Factors Replicate, but Decay after Publication

Replication rates, US (Figure 1)

Hou, Xue & Zhang (2020), raw returns	35.0%
Jensen, Kelly & Pedersen, raw returns	55.6%
Raw returns, 119 originally significant	61.3%
CAPM alpha, OLS	82.4%
CAPM alpha, Benjamini–Yekutieli	75.6%
CAPM alpha, empirical Bayes	82.4%
World, empirical Bayes	82.4%

- Steps from Hou, Xue & Zhang (2020): longer sample, 15 added factors and their construction (35.0% → 55.6%); dropping the factors never significant in the original studies (→ 61.3%); CAPM alphas instead of raw returns, the largest step (61.3% → 82.4%)
- Empirical Bayes equals OLS (82.4%): shrinkage towards zero removes discoveries, pooled precision adds them, and the two offset on average (Introduction)
- Bayesian FDR 0.1%: shrinkage controls false discoveries without the power loss of a higher t hurdle
- **Decay** after the original sample
  - mean alpha 0.49% a month in sample, 0.26% afterwards (−47%)
  - slope of post-sample on in-sample alphas 0.26 (Figure 8), against about 0.9 from shrinkage alone
- The strongest factors decay more: arbitrage or data mining (Section 12)
- Complete results: Appendix

Do 15 US Factors Replicate? (1/2)



MFM_ch3_jkp_replication

Do 15 US Factors Replicate? (2/2)

- 15 US long–short factors from value-weighted sorts of the Kenneth French Data Library, July 1963 – July 2026 ($T = 757$); CAPM alphas scaled to 10% idiosyncratic volatility; themes of Jensen, Kelly & Pedersen (authors' cluster labels)
- Replication rates: OLS 73.3% (11 of 15), Benjamini–Yekutieli 53.3% (8; critical $|t| = 2.65$), empirical Bayes 86.7% (13); details: Appendix
- Denominator: all 15 signals; whether each was significant in its original study is not checked here, unlike the 119 of the paper; replication target of Seminar 3, problem C3
- Shrinkage towards the theme, not only towards zero: long-term reversal (investment theme) 0.13% $\rightarrow$ 0.23% a month ($t = 1.25$, posterior $z = 2.85$)
- Failures: market beta and short-term reversal ($z \leq 1.41$); size only just passes ($z = 1.96$), echoing the paper, where size is one of the two themes below 50%

MFM_ch3_jkp_replication

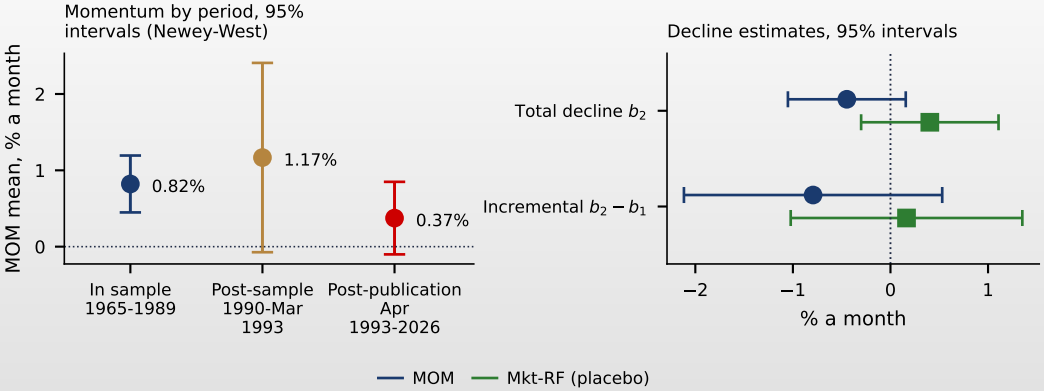
What the Course Replication Cannot Show

- **Limits of the course version**
  - 15 US signals instead of 153 in 93 countries
  - sort breakpoints of the Kenneth French Data Library instead of non-micro terciles with capped weights
  - one region, so no signal level: $\tau_s = 0$ (τ_s: dispersion of the signal components s_n)
  - the rate depends on the theme structure: with a single common theme the empirical-Bayes rate is 100%
- **Limits stated by the authors** (Appendix)
  - CAPM as the only benchmark; uniform construction of the characteristics; unobserved factors in the prior
- **Open questions**: the paper's data end in 2020
  - did the posterior alphas and the replication rate hold in 2021–2026?
  - can a prior estimated on US factors sharpen short-sample BVB factors?
- Data and code of the paper: jkpfactors.com; course project: Seminar 3, C3

Open Questions and the Discovery Loop

- **Open questions**
  - why do factor premia decay after publication: arbitrage, data mining or a change in risk?
  - does the decay differ across factors and markets?
- Evidence so far
  - McLean & Pontiff (2016): across 97 predictors, returns are 26% lower out of sample and 58% lower after publication; the gap is attributed to publication-informed trading
  - Jensen, Kelly & Pedersen (2023): most factors replicate, but the strongest decay most; Chen & Zimmermann (2022): open data and code for the published predictors
- **One pass of the loop**
  - The assistant drafts: a literature table with DOIs (digital object identifiers), competing hypotheses with their predicted patterns, the regression code, a robustness plan
  - The human fixes before running: the event dates (Jegadeesh & Titman (1993): sample ends 1989, published March 1993), the factor set, the multiple-testing correction, the placebo
  - Checks: every DOI resolves; the chapter's split-sample t-statistics are reproduced first; the data vintage is recorded
- Reusable prompts and the full checklist: ▸ Appendix

Mini-Case: Was Momentum Arbitraged Away? (1/2)



MFM_ch3_ai_discovery

Mini-Case: Was Momentum Arbitraged Away? (2/2)

- Design of McLean & Pontiff (2016): $\text{MOM}_t = a + b_1 \text{PostSample}_t + b_2 \text{PostPub}_t + e_t$, monthly, January 1965 – August 2026 ($T = 740$; the momentum file of the vintage used has August 2026, one month after the other monthly series), Newey–West errors; post-sample: January 1990 – March 1993; post-publication: from April 1993
- PostSample_t, PostPub_t: dummies equal to 1 in those periods; a: the in-sample mean; b_2: total decline relative to the in-sample mean; $b_2 - b_1$: incremental effect of publication
- $a = 0.82\%$ a month ($t = 4.32$); $b_2 = -0.45\%$ ($t = -1.45$, interval $[-1.05\%, 0.16\%]$): 54% lower; $b_2 - b_1 = -0.79\%$ ($t = -1.17$); placebo Mkt-RF $b_2 = +0.40\%$ ($t = 1.12$)
- Close to the 58% of McLean & Pontiff (2016), but one factor cannot reject $H_0 : b_2 = 0$ at 5%; arbitrage is one interpretation, not an identified cause

MFM_ch3_ai_discovery

Project Design: A Factor Discovery

- **Questions**
  - do the Fama–French factors and momentum decay after publication?
  - does the decay differ across factors?
- **Design**
  - pooled regression of SMB, HML, RMW, CMA and MOM on factor-specific post-sample and post-publication dummies
  - Mkt-RF as placebo; errors robust to cross-factor correlation and autocorrelation
  - testing for a larger decay where arbitrage is easier needs stock-level proxies (size, liquidity, short-sale costs)
- **Replicate first**: the split-sample t-statistics and the mini-case estimates

Project Design: Sample Dates and Deliverables

- **Deliverables**
  - first, an event table: factor, original paper with DOI, end of its sample, publication date, each read from the paper (model row: MOM, Jegadeesh & Titman (1993), sample ends 1989, published March 1993)
  - a pre-registration note (H_0, H_1, event dates from the table, factor set, corrections), committed before the results
  - code as a Quantlet: one folder with the script, its data source and a short description, as in the chapter's Quantlets
  - a report with Newey–West inference, placebo periods and power: the probability of rejecting $b_2 = 0$ at 5% if the true decline were, for example, the 58% of McLean & Pontiff (2016), by simulation or approximately $\Phi(\delta / \mathrm{se}(b_2) - 1.96)$ (δ: the assumed decline; Φ: standard Normal distribution function)
- **Logs**: prompts in `AI_USE.md`; caught errors in `AI_ERRORS.md`

Key Takeaways: An Empirical Workflow

1. **Estimate exposures** with HAC errors, report sample and proxy, shrink extreme betas (Blume, Vasicek)
2. **Test the pricing restriction jointly**: GRS measures the Sharpe-ratio gain from the test assets; the CAPM fails on the 25 size $\times$ B/M portfolios (flat SML, GRS $= 4.20$)
3. **Estimate premia** in two passes with errors robust to estimated betas and misspecification; a high R^2 on 25 portfolios is a low hurdle
4. **Guard against the zoo**: multiple-testing hurdles, spanning tests, model comparison by maximum Sharpe ratios
5. **Replicate**: latent factors and out-of-sample evidence; factor ETFs are cheap factor exposure, not alpha

⊡ Open questions

- why do premia decay after publication?
- how are assets priced when betas vary over time?
- which factors explain expected returns in small markets such as the BVB?

Check Your Understanding

Retrieval questions

- Why does the HAC standard error of a beta differ from the classical one, while the beta does not? (XLK example)
- Why can GRS fail to reject the CAPM on sector ETFs but reject it on the 25 portfolios over the same months? (test assets)
- Why does a useless factor look priced in pass 2? (useless factors)
- Why can the empirical-Bayes replication rate exceed the Benjamini–Yekutieli rate? (case study)

Where the ideas continue

- Chapter 2: an alpha is an anomaly or a missing risk factor, the joint hypothesis problem
- Chapter 4: factor covariance models and shrinkage for Markowitz
- Chapters 5–6: time-varying volatility, correlation and betas
- Chapter 8: backtesting factor strategies without overfitting
- Chapter 13: machine learning for the cross section of returns

GRS under the Null Hypothesis

- Under H_0 and the exact assumptions, $W \sim F(N, T-N-1)$, so $\mathbb{E}[W] = \frac{T-N-1}{T-N-3}$
- Invert the Sharpe-ratio form of W: the squared Sharpe-ratio gain expected under H_0 is

$$\mathbb{E}\left[\widehat{SR}^2(f, \mathrm{R}) - \widehat{SR}^2(f)\right] \approx \frac{N\left(1 + \widehat{SR}^2(f)\right)}{T-N-3}$$

- 25 portfolios, $T = 757$: $25 \times 1.018/729 = 0.035$ (monthly), against $0.405^2 - 0.134^2 = 0.146$ observed
- In-sample maximisation over N assets always adds some Sharpe ratio: the test asks whether the gain exceeds this noise level
- 9 sector ETFs, $T = 331$: observed gain $0.191^2 - 0.146^2 = 0.015$, below the expected $9 \times 1.021/319 = 0.029$

Back

Tests with Many Assets

- $\hat{\Sigma} = \hat{E}^\top \hat{E}/T$ has rank at most $T - 2$ (two parameters per regression); $\hat{E}$: the $T \times N$ matrix of residuals
  - Example: 100 size $\times$ B/M portfolios on 8 years of monthly data ($T = 96$): $\hat{\Sigma}$ is singular, GRS does not exist (Seminar 3, problem B7)
  - Well before this point the inverse is badly estimated and power falls
- Tests for many assets that avoid $\hat{\Sigma}^{-1}$: standardised sums of squared t-statistics, Pesaran & Yamagata (2024)
- Power enhancement by screening large alphas: Fan, Liao & Yao (2015)

Back

Misspecification-Robust Standard Errors

- Fama–MacBeth and Shanken errors assume a **correct** model: all pricing errors $e = \mu - X\gamma$ are zero ($X = [1, B]$)
- GRS rejects the model here, so γ is a pseudo-true value; the noise in $\hat{B}$ multiplies e and adds a variance term (Kan, Robotti & Shanken (2013))
- Write both passes as one exactly identified GMM system (GMM in Brief): moments $(R_t - a - Bf_t) \otimes (1, f_t)$ and $X^\top(R_t - X\gamma)$; the sandwich variance with the full Jacobian is the robust one
  - Setting $e = 0$ in the Jacobian gives the correct-model, heteroskedasticity-robust version of Jagannathan & Wang (1998)
- FF5 on the 25 portfolios (RMW: Worked Example slide): s.e. of CMA 2.2% $\to$ 2.7%, intercept 3.3% $\to$ 3.8%; HML changes by less than 0.05 percentage points
- Advanced course: Generalized Method of Moments

Back

Hansen–Jagannathan Distance and Bayesian Selection

- **Hansen–Jagannathan distance** (Hansen & Jagannathan (1997)): least-squares distance of the model SDF from the set of valid SDFs

$$\delta^2 = \min_{\mathsf{b}} \; \mathrm{e}(\mathsf{b})^\top \, \mathbb{E}[\mathrm{R}^e \mathrm{R}^{e\top}]^{-1} \, \mathrm{e}(\mathsf{b}), \qquad \mathrm{e}(\mathsf{b}) = \mathbb{E}[m_t(\mathsf{b}) \, \mathrm{R}_t^e]$$

  - Excess returns R_t^e with the normalisation $\mathbb{E}[m] = 1$; for gross returns R_t: $\mathrm{e} = \mathbb{E}[m\mathrm{R}] - 1$, and the SDF needs a free scale ($\mathbb{E}[m] = 1/(1 + R_f)$)
  - Same weighting matrix for every model, so distances are comparable; the test statistic has a weighted χ^2 limit
- Zoo-scale selection: Bayesian averaging over about 2×10^{15} (two quadrillion) factor models, robust to weak factors (Bryzgalova, Huang & Julliard (2023))
- SDF shrinkage on characteristic-based factors: a few principal components of anomaly returns capture the SDF (Kozak, Nagel & Santosh (2020))
- Common lesson: compare models by a single, model-independent metric, with its sampling uncertainty

Back

Taming the Zoo with Double-Selection LASSO

- Feng, Giglio & Xiu (2020): does a new factor add information given the existing zoo?
- **Double-selection LASSO** (Least Absolute Shrinkage and Selection Operator)
  - Step 1: a LASSO of average test-asset returns on their covariances with hundreds of factors selects the controls that price the cross section
  - Step 2: a second LASSO selects the factors whose covariances explain those of the new factor, so that omitted controls do not bias its SDF coefficient
  - Step 3: the SDF coefficient of the new factor is estimated with the union of both sets of controls and tested
- Result: most new factors are redundant; only a few add pricing information
- The case study reaches a compatible conclusion: the zoo is highly redundant, and a few themes carry the information

Back

Why the Factor-Number Estimators Disagree

- Onatski (2010): separates factor eigenvalues from the edge of the noise eigenvalue distribution
- Kaiser ($\ell_k > 1$) on a correlation matrix: with N assets and pure noise the eigenvalues spread around 1 with a width that depends on N/T
- Bai–Ng with $N = 11$ and $T = 2073$: the penalty is $\frac{N+T}{NT} \ln 11 = 0.219$ per factor
- On the sectors, $IC_{p2}(k)$ for $k = 0, \ldots, 5$ is 0, -0.86, -0.97, -1.12, -1.14, -1.17: it falls all the way to $k_{\max} = 5$, so the penalty does not stop weak factors from being added
- Report the range of estimates and the sensitivity to $k_{\max}$, not a single number (Seminar 3, problem B3)

Back

Risk-Premium PCA and Three-Pass Premia

- **RP-PCA** (Risk-Premium PCA, Lettau & Pelger (2020)): principal components of $\frac{1}{T}\mathbf{X}^\top\mathbf{X} + \gamma\bar{\mathbf{x}}\bar{\mathbf{x}}^\top$
  - $\mathbf{X}$: the $T \times N$ matrix of excess returns, $\bar{\mathbf{x}}$: their means, γ: the weight on average returns
  - $\gamma = -1$ gives covariance PCA; $\gamma > -1$ also rewards factors that explain average returns
  - In their data it finds factors with higher Sharpe ratios and smaller pricing errors than PCA
- **Three-pass estimator** (Giglio & Xiu (2021)): the premium of an observed factor g_t when the model omits factors
  - (1) PCA of test-asset returns; (2) cross-sectional regression of means on the latent betas; (3) regress g_t on the latent factors and map their premia to g_t
- Both keep PCA's statistical backbone but target pricing, not only variance; IPCA adds characteristics

Back

IPCA and Crypto-Asset Factors

- IPCA (Kelly, Pruitt & Su (2019)): characteristics predict returns because they measure risk exposure
- IPCA links PCA (statistics) with characteristic sorts (Fama–French): time-varying betas, few factors
- Crypto-assets: market, size and momentum factors price the cross section of coins (Liu, Tsyvinski & Wu (2022)); dimension reduction on crypto and traditional return features: Pele et al. (2023), Jheng et al. (2026)
- Advanced course: Multivariate Statistical Analysis

Back

Building the 153 Factors

- **Data** (Section II): 153 factors in 93 countries
  - 138 from the list of Hou, Xue & Zhang (2020), plus 15 from the wider literature
  - US returns from 1926, most developed markets from 1986
  - all in US dollars, in excess of the US Treasury bill
- **Construction**, each country and month
  - terciles of the characteristic, breakpoints from non-micro stocks
  - "capped" value weights: market equity winsorised at the NYSE 80th percentile
  - one-month holding period
  - at least 5 stocks per leg and 60 monthly observations
- **Themes**: 13 clusters of US factors (Section II.B)
  - hierarchical clustering, distance 1− correlation of CAPM residuals, Ward linkage

Back

The Bayesian Hierarchy of Alphas

- **Model** (Section I): $f_{i,t} = \alpha_i + \beta_i r_t^m + \varepsilon_{i,t}$
  - alphas $\alpha_i = c_j + s_n + w_i$ (theme j, signal n, factor i; eq. 21), prior mean 0
  - prior variance $\mathbf{\Omega} = \mathrm{M}\mathrm{M}^\top \tau_c^2 + \mathrm{Z}\mathrm{Z}^\top \tau_s^2 + \mathrm{I}\tau_w^2$ (eq. 23); posterior mean $(\mathbf{\Omega}^{-1} + T\mathbf{\Sigma}^{-1})^{-1} T\mathbf{\Sigma}^{-1}\hat{\boldsymbol{\alpha}}$ (Proposition 4)
  - M, Z: 0/1 matrices linking each factor to its theme and signal; $\tau_c^2, \tau_s^2, \tau_w^2$: variances of the three alpha components; $\mathbf{\Sigma}$: residual covariance
  - empirical Bayes: τ_c, τ_s, τ_w by maximum likelihood of $\hat{\boldsymbol{\alpha}} \sim N(0, \mathbf{\Omega} + \hat{\mathbf{\Sigma}}/T)$ (Section I.C.2)
- **Replication rate**
  - share of the factors significant in the original studies whose alpha is significant in the new data

Back

Complete Results

- Raw returns, restricted to the 119 factors significant in the original studies: 61.3% (Figure 1)
- OLS: 98 of 119 US factors; Benjamini–Yekutieli needs $|t| > 2.7$ (Figure 4)
- Bayesian FDR 0.1%, 95% interval $[0, 1\%]$; expected share of true factors 94% (Section III.C)
- Mega and large stocks: 77.3% and 79.8% (Figure 5); 11 of 13 themes at least 50%, all but low leverage and size: not an artefact of micro-caps
- Tangency Sharpe ratio (monthly), US 1972–2020: market 0.13, all factors 0.80 (Figure 14)

Back

The Course Replication in Detail

- Benjamini–Yekutieli is the strictest, as in the US rates of the paper (Figure 1)
- Net share issues move towards their theme: 0.44% $\rightarrow$ 0.27% a month
- Posterior s.d. 0.07%–0.10% against OLS s.e. 0.11%: each theme pools the evidence of its members
- **Prior dispersion**: $\hat{\tau}_c = 0.28\%$, $\hat{\tau}_w = 0.04\%$ a month
  - against 0.35% and 0.21% in Table I; 15 signals leave little factor-specific dispersion
- **Themes**: the 15 signals fall into 8 of the paper's 13 themes
  - 5 in value, 3 in low risk, 2 in investment

Back

Limits Stated by the Authors and Extensions

- **Limits stated by the authors**
  - CAPM is the only benchmark: low-leverage factors fail on CAPM alpha but matter jointly with other themes (Section III.A)
  - Characteristics are built consistently, sometimes unlike the original paper (Section II.C)
  - Unobserved factors bias the prior: with the scenarios of Harvey, Liu & Zhu (2016) the US rate falls to 81.5% and 79.8% (Figure 10)
- **Complete the course replication** (Seminar 3, C3)
  - justify each sign from the original paper
  - compare the dendrogram with the paper's themes
  - add the Bayesian FDR and the post-sample slope (Figure 8)
- Data and code of the paper: jkpfactors.com, github.com/bkelly-lab/ReplicationCrisis

Back

The Discovery Loop, Reusable Prompts (1/2)

- **Literature**
  - `Summarise the evidence since 2016 on post-publication decay and replication of anomaly returns. For each paper: DOI, sample, number of factors, headline estimate, identification of the decay. Flag entries you are unsure of.`
  - Resolve every DOI on Crossref; check each headline number against the abstract
- **Hypothesis**
  - `Contrast data mining, publication-informed arbitrage and time-varying risk as explanations of decay. For each, describe the predicted pattern across the in-sample, post-sample and post-publication periods, and across high vs low limits-to-arbitrage stocks.`
- **Data and code**
  - `Load the monthly momentum factor from the Kenneth French Data Library; from 1965, regress MOM on a post-sample dummy (1990-01 to 1993-03) and a post-publication dummy (from 1993-04), with Newey-West errors.`
  - Replicate a known number first: momentum $t = 5.53$ before 2000 and 0.75 after (split-sample chart)

Back

The Discovery Loop, Reusable Prompts (2/2)

- **Robustness**
  - `Write a pre-registered robustness plan: one publication date per factor, a Holm correction across factors, excluding the 2009 momentum crash, the working-paper date instead of the journal date, and 1927-1962 as a placebo period for Mkt-RF, SMB, HML and MOM (RMW and CMA start in July 1963).`
- **Critique**
  - `Referee this: one factor, one publication date, t = -1.45. What is the power to detect a 58% decay? Why does pooling many anomalies change it?`
- **Report**
  - In-sample mean, both dummies with Newey–West errors, the placebo, the data vintage (download date)
  - Pooled factors: errors robust to cross-factor correlation within a month and to autocorrelation
  - Prompts that shaped a decision go into `AI_USE.md`
- The assistant enumerates specifications; the human fixes which ones count before running them

Back

What the Researcher Must Check

- **Wrong event date**: sample end, working paper and journal issue are different dates
  - test: read them from the paper itself; Jegadeesh & Titman (1993): sample ends 1989, published March 1993
- **Choosing the factor after seeing it decay**: fix the set of factors in advance; Holm or BH across them
- **Serially correlated, crash-prone returns**: Newey–West errors; rerun without 2009
- **Data vintages**: the factor library is revised retroactively
  - test: record the download date; reproduce the chapter's split-sample t-statistics
- **Misquoted literature**: a common slip gives 58% as the out-of-sample decline; it is post-publication (26% out of sample)
- **Low power**: with a single factor, H_0 is rarely rejected; report the confidence interval of the decay, not only its sign

Back

Answers to the Quick Checks (1/2)

Beta, error and shrinkage

- Classical: $0.90 \pm 1.96 \times 0.05 = [0.80, 1.00]$, just below 1; Newey–West: $0.90 \pm 0.18 = [0.72, 1.08]$, contains 1: no evidence that the stock is less risky
- Report Newey–West: residuals are heteroskedastic and autocorrelated; HAC changes the estimated variance, not the OLS estimate
- $\sigma^2_\beta = 0.10 - 0.02 = 0.08$; s.e. 0.2: $w = 0.08/0.12 = 0.67$, $\beta^V = 0.67 \times 1.6 + 0.33 \times 1.0 = 1.40$; s.e. 0.1: $w = 0.08/0.09 = 0.89$, $\beta^V = 1.53$
- The noisier estimate is shrunk more; Blume gives $0.33 + 0.67 \times 1.6 = 1.40$ whatever the precision

Reading a GRS statistic

- $W = \frac{60}{4} \times (0.25^2 - 0.15^2)/(1 + 0.15^2) = 15 \times 0.040/1.0225 = 0.59 < 2.53$: not rejected
- Expected gain $4 \times 1.0225/58 = 0.071$, above the observed 0.040: the gap is within noise
- $T = 650$: $W = \frac{645}{4} \times 0.0391 = 6.31 > 2.39$: rejected; the gap is the same, the evidence grows with T

Back

Answers to the Quick Checks (2/2)

Premia, Shanken and a negative R^2

- $\text{se}_{FM} = 3\%/\sqrt{400} = 0.15\%$, $t = 3.33$
- $c = 0.5^2/1^2 = 0.25$; $\text{Var}_{\text{Sh}} = 1.25 \times (0.0225 - 0.0025) + 0.0025 = 0.0275$, s.e. 0.166%, $t = 3.02$: only just above 3
- Traded factor: $c = 0.25/4.5^2 = 0.012$, so the correction is negligible
- Mean 4%: $\text{SST} = 4 + 0 + 4 = 8$, $\text{SSE} = 16 + 0 + 16 = 32$, $R^2 = 1 - 32/8 = -3$; with a free intercept, zero slopes already give SSE = SST, and OLS can only lower SSE

Counting discoveries

- Naive 4; Bonferroni (0.01) 1; Holm: $0.004 \leq 0.01$, $0.012 \leq 0.0125$, $0.020 > 0.0167$ stop: 2; BH thresholds $0.01j$: $0.030 \leq 0.040$ is the last success: 4
- BY: the sum is 2.283, thresholds $0.0044j$: only 0.004 passes: 1
- $1 - 0.95^5 = 0.23$; a search over candidates: BH (FDR); "no factor is spurious": Holm or Bonferroni (FWER)

Back

Further Study: Courses and Code

Advanced courses on quantinar.com

- Generalized Method of Moments
- Multivariate Statistical Analysis

Code and data

- All charts and tables: github.com/danpele/MFM/tree/main/Quantlets/Ch_03
- Factor data: Kenneth French Data Library

Back

References (1/6)

- Ahn, S.C., Horenstein, A.R. (2013). Eigenvalue ratio test for the number of factors. *Econometrica*, 81(3), 1203–1227.
- Anatolyev, S., Mikusheva, A. (2022). Factor models with many assets: strong factors, weak factors, and the two-pass procedure. *Journal of Econometrics*, 229(1), 103–126.
- Ang, A., Liu, J., Schwarz, K. (2020). Using stocks or portfolios in tests of factor models. *Journal of Financial and Quantitative Analysis*, 55(3), 709–750.
- Bai, J., Ng, S. (2002). Determining the number of factors in approximate factor models. *Econometrica*, 70(1), 191–221.
- Barillas, F., Shanken, J. (2018). Comparing asset pricing models. *Journal of Finance*, 73(2), 715–754.
- Benjamini, Y., Hochberg, Y. (1995). Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society B*, 57(1), 289–300.
- Benjamini, Y., Yekutieli, D. (2001). The control of the false discovery rate in multiple testing under dependency. *Annals of Statistics*, 29(4), 1165–1188.
- Black, F. (1972). Capital market equilibrium with restricted borrowing. *Journal of Business*, 45(3), 444–455.
- Blume, M.E. (1971). On the assessment of risk. *Journal of Finance*, 26(1), 1–10.
- Bryzgalova, S., Huang, J., Julliard, C. (2023). Bayesian solutions for the factor zoo: we just ran two quadrillion models. *Journal of Finance*, 78(1), 487–557.

References (2/6)

- Campbell, J.Y., Lo, A.W., MacKinlay, A.C. (1997). The Econometrics of Financial Markets. Princeton University Press.
- Carhart, M.M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52(1), 57–82.
- Chen, A.Y., Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 11(2), 207–264.
- Cochrane, J.H. (2011). Presidential address: discount rates. *Journal of Finance*, 66(4), 1047–1108.
- Daniel, K., Moskowitz, T.J. (2016). Momentum crashes. *Journal of Financial Economics*, 122(2), 221–247.
- Fama, E.F., French, K.R. (1992). The cross-section of expected stock returns. *Journal of Finance*, 47(2), 427–465.
- Fama, E.F., French, K.R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1), 3–56.
- Fama, E.F., French, K.R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1), 1–22.
- Fama, E.F., MacBeth, J.D. (1973). Risk, return, and equilibrium: empirical tests. *Journal of Political Economy*, 81(3), 607–636.
- Fan, J., Liao, Y., Yao, J. (2015). Power enhancement in high-dimensional cross-sectional tests. *Econometrica*, 83(4), 1497–1541.

References (3/6)

- Feng, G., Giglio, S., Xiu, D. (2020). Taming the factor zoo: a test of new factors. *Journal of Finance*, 75(3), 1327–1370.
- Franke, J., Härdle, W.K., Hafner, C.M. (2019). Statistics of Financial Markets: An Introduction, 5th ed. Springer.
- Frazzini, A., Pedersen, L.H. (2014). Betting against beta. *Journal of Financial Economics*, 111(1), 1–25.
- Gibbons, M.R., Ross, S.A., Shanken, J. (1989). A test of the efficiency of a given portfolio. *Econometrica*, 57(5), 1121–1152.
- Giglio, S., Xiu, D. (2021). Asset pricing with omitted factors. *Journal of Political Economy*, 129(7), 1947–1990.
- Hansen, L.P. (1982). Large sample properties of generalized method of moments estimators. *Econometrica*, 50(4), 1029–1054.
- Hansen, L.P., Jagannathan, R. (1997). Assessing specification errors in stochastic discount factor models. *Journal of Finance*, 52(2), 557–590.
- Härdle, W.K., Simar, L. (2019). Applied Multivariate Statistical Analysis, 5th ed. Springer.
- Harvey, C.R., Liu, Y., Zhu, H. (2016). . . . and the cross-section of expected returns. *Review of Financial Studies*, 29(1), 5–68.
- Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2), 65–70.

References (4/6)

- Hou, K., Xue, C., Zhang, L. (2015). Digesting anomalies: an investment approach. *Review of Financial Studies*, 28(3), 650–705.
- Hou, K., Xue, C., Zhang, L. (2020). Replicating anomalies. *Review of Financial Studies*, 33(5), 2019–2133.
- Jagannathan, R., Wang, Z. (1998). An asymptotic theory for estimating beta-pricing models using cross-sectional regression. *Journal of Finance*, 53(4), 1285–1309.
- Jegadeesh, N., Titman, S. (1993). Returns to buying winners and selling losers: implications for stock market efficiency. *Journal of Finance*, 48(1), 65–91.
- Jensen, M.C. (1968). The performance of mutual funds in the period 1945–1964. *Journal of Finance*, 23(2), 389–416.
- Jensen, T.I., Kelly, B., Pedersen, L.H. (2023). Is there a replication crisis in finance? *Journal of Finance*, 78(5), 2465–2518.
- Jheng, S.-L., Găman, Ș., Conda, A., Pele, D.T., Härdle, W.K., Teng, H.-W., Osterrieder, J. (2026). Crypto currency returns. SSRN Working Paper.
- Kaiser, H.F. (1960). The application of electronic computers to factor analysis. *Educational and Psychological Measurement*, 20(1), 141–151.
- Kan, R., Robotti, C., Shanken, J. (2013). Pricing model performance and the two-pass cross-sectional regression methodology. *Journal of Finance*, 68(6), 2617–2649.
- Kan, R., Zhang, C. (1999). Two-pass tests of asset pricing models with useless factors. *Journal of Finance*, 54(1), 203–235.
- Kelly, B.T., Pruitt, S., Su, Y. (2019). Characteristics are covariances: a unified model of risk and return. *Journal of Financial Economics*, 134(3), 501–524.

References (5/6)

- Kleibergen, F. (2009). Tests of risk premia in linear factor models. *Journal of Econometrics*, 149(2), 149–173.
- Kozak, S., Nagel, S., Santosh, S. (2020). Shrinking the cross-section. *Journal of Financial Economics*, 135(2), 271–292.
- Lettau, M., Pelger, M. (2020). Factors that fit the time series and cross-section of stock returns. *Review of Financial Studies*, 33(5), 2274–2325.
- Lewellen, J., Nagel, S., Shanken, J. (2010). A skeptical appraisal of asset pricing tests. *Journal of Financial Economics*, 96(2), 175–194.
- Lintner, J. (1965). The valuation of risk assets and the selection of risky investments in stock portfolios and capital budgets. *Review of Economics and Statistics*, 47(1), 13–37.
- Liu, Y., Tsyvinski, A., Wu, X. (2022). Common risk factors in cryptocurrency. *Journal of Finance*, 77(2), 1133–1177.
- Markowitz, H. (1952). Portfolio selection. *Journal of Finance*, 7(1), 77–91.
- McLean, R.D., Pontiff, J. (2016). Does academic research destroy stock return predictability? *Journal of Finance*, 71(1), 5–32.
- Newey, W.K., West, K.D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.
- Newey, W.K., West, K.D. (1994). Automatic lag selection in covariance matrix estimation. *Review of Economic Studies*, 61(4), 631–653.
- Onatski, A. (2010). Determining the number of factors from empirical distribution of eigenvalues. *Review of Economics and Statistics*, 92(4), 1004–1016.

References (6/6)

- Pele, D.T., Wesselhöfft, N., Härdle, W.K., Kolossiatis, M., Yatracos, Y.G. (2023). Are cryptos becoming alternative assets? *The European Journal of Finance*, 29(10), 1064–1105.
- Pesaran, M.H., Yamagata, T. (2024). Testing for alpha in linear factor pricing models with a large number of securities. *Journal of Financial Econometrics*, 22(2), 407–460.
- Politis, D.N., Romano, J.P. (1994). The stationary bootstrap. *Journal of the American Statistical Association*, 89(428), 1303–1313.
- Roll, R. (1977). A critique of the asset pricing theory's tests. Part I. *Journal of Financial Economics*, 4(2), 129–176.
- Ross, S.A. (1976). The arbitrage theory of capital asset pricing. *Journal of Economic Theory*, 13(3), 341–360.
- Shanken, J. (1992). On the estimation of beta-pricing models. *Review of Financial Studies*, 5(1), 1–33.
- Sharpe, W.F. (1964). Capital asset prices: a theory of market equilibrium under conditions of risk. *Journal of Finance*, 19(3), 425–442.
- Vasicek, O.A. (1973). A note on using cross-sectional information in Bayesian estimation of security betas. *Journal of Finance*, 28(5), 1233–1239.
- White, H. (1980). A heteroskedasticity-consistent covariance matrix estimator and a direct test for heteroskedasticity. *Econometrica*, 48(4), 817–838.
- Data: Kenneth R. French Data Library, mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.

Acronyms used in this material (1/2)

- **AI**: Artificial Intelligence
- **APT**: Arbitrage Pricing Theory
- **BAB**: Betting Against Beta
- **BET**: Bucharest Exchange Trading (index)
- **BH**: Benjamini–Hochberg (procedure)
- **BVB**: Bursa de Valori București — *the Bucharest Stock Exchange*
- **CAGR**: Compound Annual Growth Rate
- **CAPM**: Capital Asset Pricing Model
- **CMA**: Conservative Minus Aggressive (investment factor)
- **CML**: Capital Market Line
- **ETF**: Exchange-Traded Fund
- **FDR**: False Discovery Rate
- **FF3**: Fama–French three-factor model
- **FF5**: Fama–French five-factor model
- **FWER**: Family-Wise Error Rate
- **GLS**: Generalised Least Squares
- **GMM**: Generalized Method of Moments
- **GRS**: Gibbons–Ross–Shanken (test)
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HML**: High Minus Low (value factor)

Acronyms used in this material (2/2)

- **IPCA**: Instrumented Principal Component Analysis
- **LASSO**: Least Absolute Shrinkage and Selection Operator
- **MOM**: Momentum (factor)
- **NYSE**: New York Stock Exchange
- **OLS**: Ordinary Least Squares
- **OMV**: Österreichische Mineralölverwaltung — *Austrian Mineral Oil Administration (OMV group)*
- **PC1**: first Principal Component
- **PC2**: second Principal Component
- **PC3**: third Principal Component
- **PCA**: Principal Component Analysis
- **PRDS**: Positive Regression Dependence on a Subset (of p-values)
- **RF**: Risk-Free rate
- **RMSE**: Root Mean Squared Error
- **RMW**: Robust Minus Weak (profitability factor)
- **ROE**: Return On Equity
- **RP-PCA**: Risk-Premium Principal Component Analysis
- **SDF**: Stochastic Discount Factor
- **SMB**: Small Minus Big (size factor)
- **SML**: Security Market Line
- **UMD**: Up Minus Down (momentum factor)
- **US**: United States